

**VISTA: A NLP-BASED CHATBOT FOR CITIZEN INQUIRY
AND LOCAL GOVERNMENT SERVICES**



**A Capstone Project Presented to
Faculty of the Institute of Computing
Davao del Norte State College
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CHAPTER 1

INTRODUCTION

Background of the Project

The rapid advancement of digital technologies has fundamentally transformed how governments deliver public services and communicate with citizens across the globe. Public institutions worldwide are adopting electronic government (e-government) initiatives to improve service efficiency, transparency, and accessibility through information and communication technologies (ICT) [1]. Digital platforms allow citizens to access official information remotely, reducing the need for physical visits and helping standardize service communication across channels. Research highlights that chatbots, a conversational interface capable of interpreting user questions and providing automated responses can support organizations by delivering consistent answers to frequently asked questions and reducing the workload associated with repetitive inquiries [2]. In government settings, AI-guided chatbots enhance citizen–government communication by improving responsiveness and standardizing information delivery [3]. From a governance perspective, digital platforms support "doing more with less" by enabling government offices to allocate human resources to complex, judgment-based tasks while routine information delivery is supported by technology [4].

The core problem identified in this study is the inefficient and fragmented handling of citizen inquiries at the local government level, specifically in Panabo City, Davao del Norte. Citizens commonly inquire about frontline services documented in the

Citizen's Charter, including business permitting, civil registry, and local treasury services. Although the Citizen's Charter serves as an official reference, accessing information can still be inconvenient for citizens when it is not easily searchable in natural language. The root causes include the absence of a centralized automated response system, limited personnel capacity during peak inquiry periods, the lack of a 24/7 accessible digital inquiry platform, and the absence of multilingual support for local languages such as Cebuano/Bisaya and Filipino/Tagalog. These conditions produce cascading effects: frustrated citizens unable to access timely information, delayed and inconsistent dissemination of service requirements, overwhelmed frontline staff burdened by high volumes of repetitive inquiries, and reduced public trust in LGU responsiveness.

Globally, one of the persistent challenges in public service delivery is the large volume of repetitive citizen inquiries, particularly questions about service requirements, documentary needs, step-by-step procedures, office schedules, fees, and processing timelines. According to the United Nations E-Government Survey 2022, digital government initiatives are critical for strengthening institutional transparency, improving public service delivery, and fostering citizen participation in governance processes [1]. Several international studies demonstrate the effectiveness of AI-driven chatbot systems in government service delivery. Androutsopoulou et al. [3] showed how AI-guided chatbots transform citizen–government communication by providing consistent, round-the-clock responses to municipal service inquiries, significantly reducing the burden on frontline staff. [Khurana et al. \[2\]](#) further highlights how Natural Language Processing techniques enable machines to interpret the semantic meaning of

human text, making NLP the foundational technology for any modern chatbot system. Adamopoulou and Moussiades [5] provide a comprehensive historical and technical survey of chatbot evolution, establishing that retrieval-based chatbots which match user inputs against a predefined knowledge base are the most suitable architecture for domains requiring verified, accurate responses, such as government services.

In the Philippines, national programs continue to promote digital transformation through ICT-enabled initiatives, as outlined in the Department of Information and Communications Technology (DICT) E-Government Masterplan 2022 [6]. To streamline services, Republic Act No. 11032, also known as the Ease of Doing Business and Efficient Government Service Delivery Act, mandates all local government units (LGUs) to establish a Citizen's Charter detailing step-by-step procedures, requirements, fees, and processing times for all frontline services [7]. Despite these policies, many LGUs still face significant gaps in implementing centralized, automated inquiry systems, with most relying on basic social media auto-replies or manual front-desk handling [8]. The Anti-Red Tape Authority (ARTA) further reinforces the Citizen's Charter requirement through reference guidelines that standardize how government agencies should present service information to the public [9], yet localized digital application in municipal and city governments remains a significant challenge due to resource constraints. Furthermore, Filipino and Cebuano are classified as low-resource languages in NLP research, meaning there is a critical scarcity of publicly available, labeled datasets for tasks such as intent classification and text categorization in these languages [10].

At the local level, existing digital inquiry solutions, where available, remain largely inaccessible to local government units due to their reliance on proprietary platforms,

computationally heavy deep learning models, or English-only interfaces, leaving a critical gap between nationally available technologies and the operational realities of city-level government offices in linguistically diverse regions like Mindanao. In Panabo City, a first-class component city in Davao del Norte with a population of 184,599 as of the 2020 Census [11], citizen inquiries are still handled through manual and fragmented channels such as front-desk assistance, phone calls, and social media messaging. While these channels remain important, they may lead to delayed responses during peak hours, inconsistent answers depending on staff availability, and inefficiencies for both citizens and government personnel.

This study addresses that gap by developing VISTA (Virtual Intelligent Services and Transactions Assistant): An NLP-based Chatbot for Citizen Inquiry and Local Government Services. VISTA is a knowledge-based, retrieval-oriented chatbot that integrates Natural Language Processing (NLP) with supervised machine learning algorithms to classify citizen inquiries and retrieve validated responses from the LGU's Citizen's Charter [5]. The system supports multilingual inquiries in English, Filipino/Tagalog, and Cebuano/Bisaya, and features a controlled retraining mechanism where only administrator-validated queries are incorporated into the model, ensuring that the chatbot provides only verified, official information. Conducted under the research, development, and extension mandate of Davao del Norte State College as provided under Republic Act 7879 [12], this study aims to deliver an accessible, locally deployable, and technically reliable solution tailored to the needs of citizens and frontline offices in Panabo City.

This study is aligned with several national and international development agendas. At the institutional level, the project supports the Davao del Norte State College Research, Development, and Extension (DNSC RDE) Agenda, which prioritizes the development of technology-driven solutions that address community-level concerns and contribute to local governance improvement [13]. At the national level, the study aligns with the Commission on Higher Education (CHED) Achieve Agenda, which encourages higher education institutions to produce research outputs that are responsive to societal needs and contribute to national development goals through innovation and technology [14]. The study also supports the Department of Science and Technology (DOST) Harmonized Research and Development Agenda, which promotes the development of smart technologies and digital platforms for improving public service delivery and governance efficiency in the Philippines [15]. At the international level, this study directly contributes to three United Nations Sustainable Development Goals (UN SDGs): SDG 9 (Industry, Innovation, and Infrastructure), specifically Target 9.c, which promotes the development of information and communication technology to increase access to the internet and digital services in developing countries [16]; SDG 11 (Sustainable Cities and Communities), specifically Target 11.3, which advocates for inclusive, participatory, and integrated planning and management of urban services; and SDG 16 (Peace, Justice, and Strong Institutions), specifically Target 16.6, which calls for the development of effective, accountable, and transparent institutions at all levels [16].

Objectives of the Study

This study aims to develop a web-based system for citizen inquiry and local government services integrating NLP-based chatbot to automate the delivery of verified public service information and streamline inquiry management. Specifically, the project aims to:

1. Develop a structured, multilingual knowledge base (English, Filipino/Tagalog, Cebuano/Bisaya) from LGU-approved sources such as the Citizen's Charter and FAQs.
2. Preprocess the textual data using NLP techniques and apply TF-IDF vectorization to convert text into numerical representations suitable for machine learning classification and response retrieval.
3. Train and evaluate three supervised machine learning algorithms to identify the most accurate intent classification model based on accuracy, precision, recall, and F1-score.
4. Design an administrator module with an analytics dashboard to manage the knowledge base, monitor inquiry trends, and implement a controlled retraining mechanism that incorporates only validated unmatched queries to improve model performance.

Significance of the Study

This study is significant as it provides an automated, real-time, and multilingual information retrieval system that transforms traditional reactive service delivery into proactive citizen inquiry management. The following are the study's main beneficiaries:

- **Citizens and Residents of Panabo City.** VISTA will enable citizens to access official service information — including requirements, procedures, fees, schedules, and processing times — instantly and in their preferred language (English, Filipino/Tagalog, or Cebuano/Bisaya) without requiring immediate office visits. The chatbot provides consistent, validated answers sourced from the Citizen's Charter, reducing waiting time, repeated visits, and the frustration of receiving inconsistent information across different inquiry channels.
- **Local Government Frontline Offices.** The system will help frontline offices of Panabo City Hall reduce the volume of repetitive inquiries handled manually, supporting the consistent dissemination of official service requirements anchored on LGU-approved references. By automating routine information delivery, VISTA allows frontline offices to operate more efficiently, particularly during peak service hours and high-demand periods.
- **Frontline Government Personnel.** By redirecting routine inquiries to the chatbot, frontline staff can prioritize case-specific concerns, complex transactions, and tasks that require human judgment. This reduces staff fatigue, minimizes the risk of providing inconsistent answers, and allows personnel to focus on value-added activities.
- **LGU Management and Planning Units.** VISTA's analytics dashboard will provide management with data-driven insights into citizen inquiry trends, including the most frequently asked service categories, peak inquiry periods, and common unresolved questions. These analytics support evidence-based

communication planning, resource allocation, and service improvement decisions.

- **Davao del Norte State College (DNSC).** The development of VISTA will strengthen DNSC's research, development, and extension (RDE) portfolio by producing an applied technological output that directly addresses a real challenge in local government service delivery, reinforcing DNSC's mandate under Republic Act 7879 [12].
- **Future Researchers and Developers.** This study lays the foundation for future research on NLP-based smart governance systems applicable to citizen service delivery in linguistically diverse regions. VISTA may serve as a baseline system for extensions such as voice interaction, sentiment analysis, and improved algorithm architectures.

Scope and Limitation

This project focuses on the development of VISTA, a web-based multilingual (English, Filipino/Tagalog, Cebuano/Bisaya) chatbot that provides automated responses to citizen inquiries about selected Panabo City Hall frontline services. The system will be tested and evaluated within Panabo City, Davao del Norte. The system encompasses four primary operational components: a citizen-facing chatbot interface for multilingual inquiry submission and response retrieval, an NLP-based intent classification module using supervised machine learning algorithms (SVM, Logistic Regression, Neural Network), an administrator portal for knowledge base management and controlled model retraining, and an analytics dashboard for inquiry logging and trend visualization.

The initial deployment and testing of the system will be limited to a baseline of approximately 1,500 pre-defined question variations distributed across five (5) selected frontline offices of Panabo City Hall. These five offices and their approximate question allocations are as follows: City Mayor's Office with approximately 1,000 question variations covering its extensive range of services, City Civil Registrar's Office with approximately 100 question variations, City Treasurer's Office with approximately 100 question variations, City Assessor's Office with approximately 150 question variations, and City Social Welfare and Development Office with approximately 150 question variations. The questions for each service consist of the most common frequently asked questions (FAQs) gathered from the Citizen's Charter and historical citizen inquiries from the LGU's public communication channels. This 1, 500-question baseline exists because it is not feasible to anticipate every possible citizen inquiry phrasing during the initial data gathering phase.

However, the system is not limited in its capacity to accept additional questions. VISTA includes a controlled retraining feature that allows the administrator to continuously expand the knowledge base beyond the initial baseline over time. When citizens submit inquiries that the system cannot match to an existing intent (unmatched queries), these queries are logged and queued for administrator review. Once the administrator reviews and resolves an unmatched query by assigning it to the correct intent or creating a new intent, the resolved query is permanently added to the knowledge base and training dataset. This mechanism serves as the system's room for improvement, it ensures that the chatbot continuously learns from real-world citizen

interactions while maintaining the accuracy and trustworthiness of its responses, since all additions require explicit human validation before being incorporated into the model.

The administrator of the system is the designated authorized LGU personnel, such as the Public Information Officer or IT staff of Panabo City Hall, who is responsible for monitoring the chatbot, resolving unmatched queries, updating the knowledge base, and triggering model retraining through the administrator portal. The system is built around a Python-based backend server integrated with a TF-IDF feature extraction pipeline and supervised machine learning classifiers that classify citizen text inquiries into predefined service intent categories in real time. Development and testing of the prototype will be conducted using service information documented in the Panabo City Citizen's Charter, and while the system is designed to be adaptable to other LGU contexts, findings and system configurations may require modification before deployment in other municipal or provincial government offices.

VISTA is limited to providing information support and will not process official government transactions, payments, or document submissions. The accuracy of the chatbot's responses is highly dependent on the quality and completeness of the manually curated knowledge base maintained by LGU administrators. NLP classification performance may vary across languages due to local slang, code-switching, and spelling variations inherent in Cebuano/Bisaya and Filipino/Tagalog text inputs. The system does not feature automatic real-time self-learning to prevent the AI from adopting incorrect or malicious user inputs; all learning requires explicit administrator review and validation. The web application requires a stable internet connection and

modern web browsers to function effectively as a prototype.

Review of Related Literature and Works

Related Literature

The integration of information and communication technologies in public administration has fundamentally shifted how governments interact with citizens. According to the United Nations E-Government Survey 2022, digital government initiatives are critical for strengthening institutional transparency, improving public service delivery, and fostering citizen participation in governance processes [1]. As these platforms mature, chatbots have emerged as a primary interface for navigating complex government portals and delivering automated responses to citizen inquiries [2]. In the public sector, studies show that AI-guided chatbots significantly reduce response times and mitigate the workload of frontline personnel who handle high volumes of repetitive inquiries [3]. This aligns with the principles of "lean government," where technology handles routine tasks, allowing human workers to focus on specialized service delivery that requires professional judgment [4].

Several studies demonstrate the effectiveness of AI-driven chatbot systems in government service delivery. Androutsopoulou et al. [3] showed how AI-guided chatbots transform citizen–government communication by providing consistent, round-the-clock responses to municipal service inquiries, significantly reducing the burden on frontline staff. The Bologna City chatbot implementation validated the use of Support Vector Machines (SVM) combined with TF-IDF feature extraction for classifying citizen inquiries into service categories with high accuracy. Similarly, research by Khurana et al. [2]

highlights how Natural Language Processing techniques enable machines to interpret the semantic meaning of human text, making NLP the foundational technology for any modern chatbot system. Adamopoulou and Moussiades [5] provide a comprehensive historical and technical survey of chatbot evolution, establishing that retrieval-based chatbots which match user inputs against a predefined knowledge base are the most suitable architecture for domains requiring verified, accurate responses, such as government services.

In the Philippine context, the push for automated, accessible e-governance is mandated by national frameworks including the DICT E-Government Masterplan 2022 [6] and the Anti-Red Tape Authority (ARTA) guidelines on the Citizen's Charter [9], yet localized application in municipal and city governments remains a significant challenge due to resource constraints [8]. Republic Act No. 11032 requires all LGUs to maintain a Citizen's Charter [7], providing a standardized, official data source that can serve as the foundation for a knowledge-based chatbot system.

A dominant theme across the literature is the reliance on Natural Language Processing to accurately classify user intentions [5]. To process text, raw user input must be converted into numerical data. Literature heavily supports the use of Term Frequency-Inverse Document Frequency (TF-IDF) for feature extraction, as it effectively weighs the importance of specific keywords in a query while filtering out common conversational noise (stop words) [17]. TF-IDF has been shown to outperform simpler bag-of-words approaches in structured domain classification tasks due to its ability to distinguish meaningful terms from generic vocabulary.

Regarding classification, the literature justifies the use of traditional machine learning algorithms over computationally heavy deep learning models for text categorization in structured domains. Support Vector Machine (SVM) is widely utilized because it excels in high-dimensional text spaces and provides robust accuracy for classification tasks, particularly when the number of features exceeds the number of training samples [18]. Logistic Regression is also favored as a reliable probabilistic model for multiclass text problems, offering interpretable probability estimates for each predicted class [19]. Neural Networks, specifically the Multilayer Perceptron (MLP), have demonstrated strong performance in text classification tasks due to their ability to learn non-linear patterns and complex feature interactions that linear models may miss [20]. By utilizing robust machine learning libraries such as Scikit-learn to evaluate these three specific algorithms [21], developers can ensure a mathematically sound, lightweight, and highly accurate intent classification pipeline suitable for deployment on resource-constrained LGU servers.

Across these studies, key themes and patterns emerged. The common findings confirm that NLP-based chatbots using supervised machine learning classifiers provide accurate, consistent, and efficient responses to structured domain inquiries. Common methodological trends include the use of TF-IDF for feature extraction, SVM and neural network-based classifiers for text classification, and retrieval-based architectures for response generation. On the other hand, most existing systems are limited to single-language (English) support and lack controlled retraining mechanisms that allow administrators to validate and approve new training data before it is incorporated into the model.

Despite the proven benefits of AI chatbots and algorithms like SVM and neural network classifiers, a significant research gap exists at the local government level. Many LGUs rely on basic social media auto-replies or manual handling [8]. Furthermore, existing advanced government chatbots are predominantly built for the English language and do not account for local Philippine dialects (Cebuano/Bisaya) or code-switching. Moreover, Filipino and Cebuano are classified as low-resource languages in NLP research, meaning there is a critical scarcity of publicly available, labeled datasets for tasks such as intent classification and text categorization in these languages [10]. This absence is even more pronounced in the government services domain, where no pre-built training corpus exists for Philippine LGU citizen inquiries — forcing system developers to construct their own datasets from primary documents such as the Citizen's Charter. Finally, many chatbot systems lack a "controlled retraining workflow," meaning they either remain static and become outdated, or they use unsupervised learning which risks providing citizens with unverified information.

The literature confirms that an NLP-based chatbot using TF-IDF and supervised learning algorithms (SVM, Logistic Regression, Neural Network) is a highly effective solution for managing routine inquiries. However, the lack of localized, multilingual systems anchored on verified documents like the Citizen's Charter highlights a clear gap. VISTA addresses this by providing a controlled, governable AI framework tailored specifically for Panabo City's linguistic and administrative context.

Related Works

There are five existing automated inquiry systems and published studies that can be compared to the proposed VISTA system. These systems utilize natural language

processing, government knowledge bases, and conversational interfaces to automate public service inquiries. However, the proposed VISTA system introduces additional critical functionalities tailored for local government units (LGUs), such as multilingual support for English, Tagalog, and Cebuano/Bisaya, a controlled retraining mechanism, an admin analytics dashboard, and an open-source, locally deployable architecture.

The first related system is the eGovPH Portal [22]. This system is primarily designed as a centralized platform for Philippine government services, providing citizens with access to a verified knowledge base of national and local government transactions. Through its recent eGovAI integration, the platform utilizes Natural Language Processing (NLP) to handle conversational inquiries and offers multilingual support for various Philippine dialects. Figure 1 shows the interfaces of the system [6].

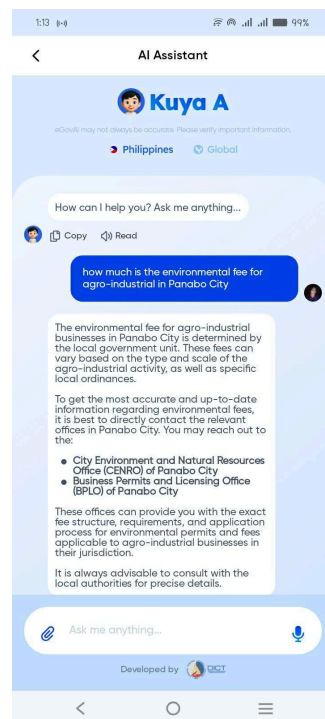


Figure 1. eGovPH Ai Assistant Interface

Although the system is highly reliable and advanced, it functions as a centralized, closed-source national platform. There is no open-source, locally deployable framework that individual Local Government Units (LGUs) can host, govern, and customize on their own servers. Furthermore, it lacks a localized, controlled retraining mechanism that would allow non-technical municipal staff to independently update their city's specific dataset (such as a localized Citizen's Charter). Thus, while eGovPH excels at the national level, it does not fully satisfy the localized, deployable criteria presented in Table 1.

The second related system is AskGov Singapore [23]. This system focuses on answering citizen queries through an AI-guided conversational interface utilizing advanced NLP models, specifically BERT, to classify user intents and retrieve information from verified government databases [3]. Its key features include accurate intent classification, automated response generation, and centralized knowledge management. The system demonstrates how modern AI improves government communication efficiency. Figure 2 below shows the interface of the system.

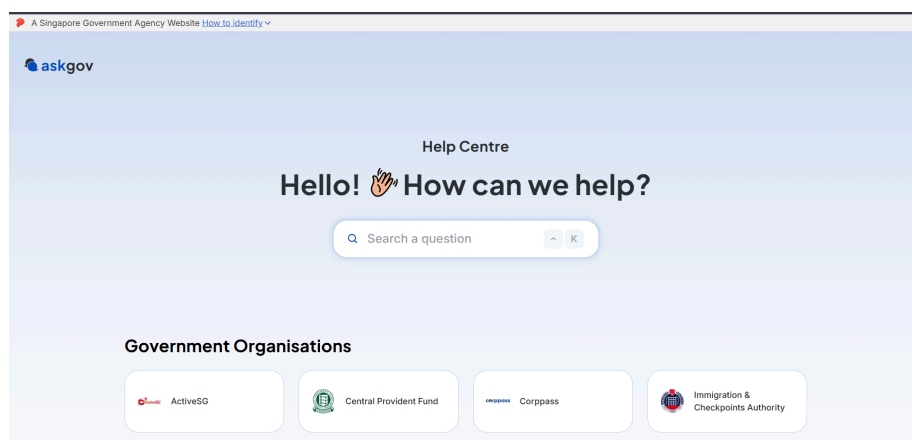


Figure 2. AskGov Conversational Interface

Although the system successfully demonstrates NLP intent classification and a verified knowledge base, it mainly focuses on single-language interactions and lacks multilingual support for Philippine regional languages (Tagalog and Cebuano/Bisaya). The system also does not specifically implement a controlled retraining mechanism where administrators manually validate unmatched queries before the model learns them. Additionally, it is a proprietary national system rather than an open-source tool deployable by local municipalities.

The third related study is the Bologna City Chatbot [24] developed for municipal citizen services. This system utilizes Support Vector Machines (SVM) combined with TF-IDF feature extraction to classify citizen text inquiries into specific municipal service categories. The system provides automated, round-the-clock responses to routine administrative questions, significantly reducing the workload on city hall staff. Figure 3 illustrates the Interface of the chatbot.

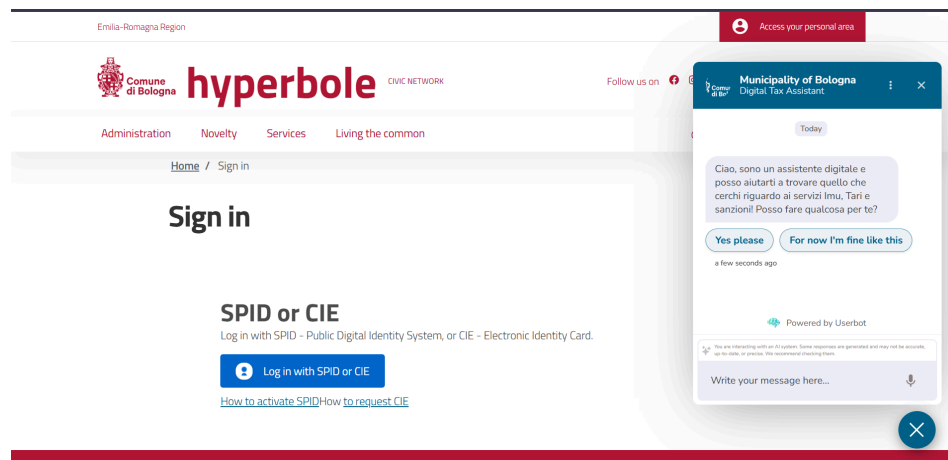


Figure 3. Bologna City Chatbot Interface

However, the study primarily focuses on European languages and does not support the linguistic nuances and code-switching common in the Philippines (English,

Tagalog, Bisaya). Furthermore, the system lacks a dedicated admin analytics dashboard for monitoring inquiry trends and a controlled retraining module for continuous human-in-the-loop knowledge base expansion.

The fourth related system is the PhilHealth Virtual Assistant [25]. This system uses Google Dialogflow to provide automated assistance to members regarding their health insurance benefits, contributions, and membership status. Its key features include NLP intent classification and predefined conversational flows. Figure 4 shows the user interface of the PhilHealth Virtual Assistant.



Figure 4. PhilHealth Virtual Assistant Interface

Although the Virtual Assistant demonstrates the effectiveness of intent classification in the Philippine public sector, its knowledge base is strictly limited to national health insurance rather than comprehensive local government services. The system does not utilize a locally deployable architecture, relying instead on proprietary cloud services (Dialogflow). It also lacks targeted support for Cebuano/Bisaya text inputs and a custom admin analytics dashboard.

The fifth related system is the Sophia Chatbot [26], an AI-powered conversational assistant adopted by the Quezon City local government to support victim-survivors of domestic violence. Developed by the non-profit organization Spring ACT, the system utilizes Natural Language Processing (NLP) to understand user inputs and securely guide citizens to appropriate legal resources, evidence collection tools, and municipal protection centers. While it successfully demonstrates AI chatbot integration within a Philippine LGU and supports local languages, its knowledge base is strictly domain-locked to domestic violence advocacy rather than comprehensive local government services like the Citizen's Charter. Additionally, because it is centrally managed on a secure global network to prioritize strict anonymity, it lacks an open-source, locally deployable framework and a controlled retraining dashboard that would allow municipal administrators to validate and expand general civic intents independently. Figure 5 shows the Interface of Sophia chatbot.

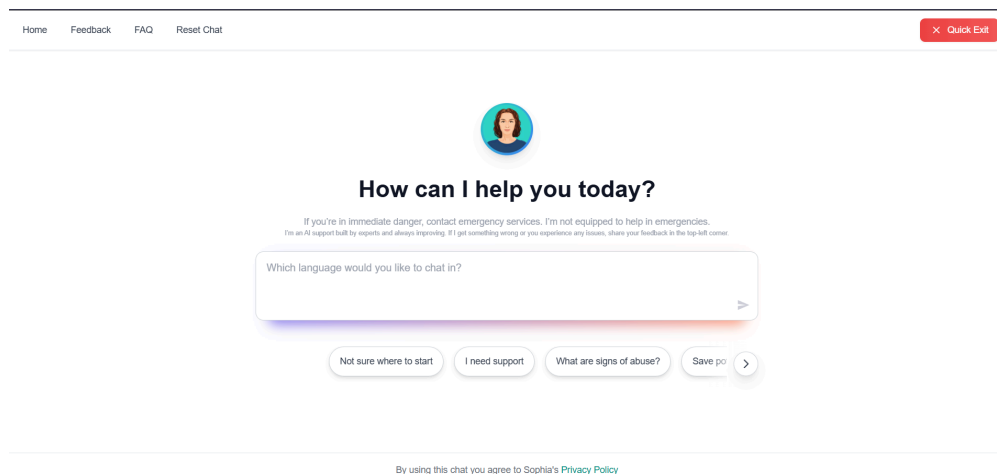


Figure 5. Sophia Chatbot Interface

Table 1 presents a comparison table evaluating VISTA alongside five existing automated inquiry systems across six key features: NLP intent classification, multilingual support, verified knowledge base, controlled retraining, admin analytics dashboard, and open-source deployability. The comparison reveals that while existing systems offer partial capabilities, none provides a complete and integrated solution that combines all six features into a single system.

Table 1. Comparison of Existing Systems and Proposed System

Title of the Study / System	NLP Intent Classification	Multilingual Support	Verified Knowledge Base	Controlled Retraining	Admin Analytics Dashboard	Open-Source / Locally Deployable
eGovPH Portal [6]			✓			
AskGov Singapore [3]	✓ (BERT)		✓			
Bologna City Chatbot [4]	✓ (SVM)		✓			
PhilHealth Virtual Assistant	✓ (Dialogflow)		Partial			

Jakarta e-Gov Bot	✓ (Naïve Bayes)		Partial			Partial
VISTA (Proposed System)	✓ (SVM/LR/MLP)	✓ (EN/TL/BS)	✓ (Citizen's Charter)	✓	✓	✓

Based on the reviewed literature and related works, existing systems provide useful features such as NLP intent classification, verified knowledge bases, and automated conversational interfaces. However, most systems do not combine multilingual support (English, Tagalog, Cebuano), controlled retraining, an admin analytics dashboard, and an open-source locally deployable architecture in one system. This gap supports the development of VISTA as a comprehensive, multilingual, and locally manageable automated inquiry system tailored for Panabo City Hall and similar local government units.

Definition of Terms

Administrator Portal	A secured web-based module used by authorized LGU personnel to manage VISTA's knowledge base, review unmatched citizen queries, update FAQ entries, and trigger controlled model retraining. In this study, it refers to the backend management interface accessible only through authenticated administrator credentials.
Analytics Dashboard	The system component that visualizes inquiry frequency, top service intents, peak inquiry periods, and unmatched query statistics based on aggregated inquiry logs. In this study, it refers to the data visualization interface within the administrator portal.
Artificial Intelligence (AI)	Computing methods that enable systems to perform tasks that typically require human intelligence, such as language interpretation, pattern recognition, and

	decision-making. In this study, AI refers specifically to the machine learning algorithms used for intent classification.
Cebuano/Bisaya	A local language widely used in Mindanao; included as a supported inquiry language in VISTA to serve the linguistic diversity of the Panabo City community.
Chatbot	A conversational system that interacts with users through text-based interfaces and provides automated responses using NLP techniques and a structured knowledge base. In this study, it refers to the citizen-facing VISTA web interface.
Citizen's Charter	An official LGU document mandated by Republic Act No. 11032, describing all frontline services, requirements, procedures, fees, and processing times. In this study, it serves as the primary validated data source for VISTA's knowledge base.
Confusion Matrix	A classification evaluation table used to compute the accuracy, precision, recall, and F1-score of the machine learning models by comparing predicted intent labels against actual intent labels.
Controlled Retraining	A supervised learning mechanism where only administrator-reviewed and validated unmatched queries are incorporated into the training dataset before triggering a model retrain cycle. This prevents the AI from adopting incorrect or malicious user inputs.
Intent Classification	The process of utilizing machine learning algorithms to assign a user's text inquiry to a specific service category (intent) in order to retrieve the appropriate validated response from the knowledge base.
Knowledge Base	A structured repository of validated service information — including requirements, procedures, schedules, fees, and processing times — sourced from the Citizen's Charter and used for automated response retrieval.
Logistic Regression	A probabilistic machine learning algorithm used for multiclass text classification that estimates the probability that a given input belongs to each possible intent category.
Multilayer Perceptron (MLP)	A type of artificial neural network consisting of an input layer, one or more hidden layers, and an output layer that learns non-linear patterns in text features through backpropagation, used as one of three candidate algorithms for VISTA's intent classification pipeline.

Natural Language Processing (NLP)	A subfield of artificial intelligence that provides techniques enabling computers to process, clean, and interpret human language text for automated understanding and response generation.
Stop Words	Common conversational words (e.g., "the," "is," "and," "ang," "sa") that are removed during text preprocessing to improve the algorithm's focus on meaningful, domain-specific key terms.
Support Vector Machine (SVM)	A supervised machine learning algorithm that creates an optimal hyperplane to separate different intent categories in high-dimensional feature space, used as one of three candidate algorithms for VISTA's intent classification pipeline.
TF-IDF (Term Frequency–Inverse Document Frequency)	A statistical method used to convert text into numerical feature vectors by evaluating how relevant a word is to a document within a collection of documents, giving higher weight to distinctive keywords and lower weight to common words.
VISTA	Virtual Intelligent Services and Transactions Assistant; the name of the proposed NLP-based chatbot system designed to automate citizen inquiry handling for Panabo City local government services.
Confidence Threshold	The minimum probability score (set at 35%) that the machine learning classifier must produce for a predicted intent before the system considers the classification valid and returns a response.
Intent	A predefined category representing a specific citizen service inquiry (e.g., "business_permit_fee", "birth_certificate_process") used by the classifier to route queries to the correct knowledge base response

CHAPTER 2

METHODOLOGY

The methodology used in the development of VISTA, an NLP-based chatbot for citizen inquiry and local government services, is discussed in this chapter. It covers the overall development process of the web-based chatbot system, including the chosen System Development Life Cycle (SDLC) model, the PADIM framework, and the specific step-by-step process used in planning, analysis, design, implementation, testing, and maintenance of the system. This chapter also outlines the tools, diagrams, requirements, and processes involved in ensuring the success of this proposed project.

VISTA adopts an Agile SDLC model, which is essential for web-based systems that integrate machine learning pipelines, as it requires continuous improvement, iterative testing, and incremental development. This integration enables the balance between structure and adaptability, making it suitable for NLP-based systems with both well-defined requirements and evolving components such as multilingual training datasets and classification model accuracy [27]. The Agile framework shown in figure 6 that is used in this study follows the PADIM cycle: Planning, Analysis, Design, Implementation, and Maintenance. Each phase is executed iteratively, allowing the development team to revisit and refine components based on testing feedback and evolving requirements.

In the Planning phase, the proponents defined the project scope, objectives, timeline, and required resources based on the approved concept paper. During the Analysis phase, the proponents gathered and finalized system requirements by

reviewing existing LGU inquiry processes, identifying citizen needs, and determining both functional and non-functional requirements. In the Design phase, the proponents translated these requirements into a structured system architecture, including the design of the database, user interface, NLP classification pipeline, and overall system flow, supported by diagrams and prototypes. During the Implementation phase, the proponents developed the system using an iterative approach, where coding, integration of NLP components, and testing were performed incrementally, allowing them to refine features such as intent classification accuracy, multilingual support, and controlled retraining mechanisms based on testing results. Finally, in the Maintenance phase, the proponents conducted system evaluation, identified issues, and applied necessary improvements to enhance system performance and usability.

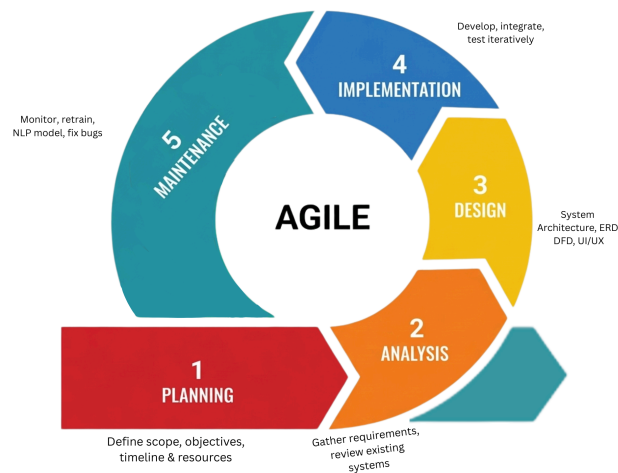


Figure 6. PADIM / Agile SDLC Cycle

System Planning

Systems Planning sets the groundwork for VISTA by outlining the goals, scope, feasibility, resources, risks, and timeline of the project. Major activities involve identifying the needs of citizens and LGU administrators in receiving prompt and accurate

responses to frontline service inquiries. The researchers will define the main objective of the system, which is to develop a web-based, multilingual chatbot powered by Natural Language Processing (NLP) that can accurately classify citizen inquiries and retrieve validated responses from the knowledge base through a centralized interface. A feasibility appraisal will be conducted to assess the technical and operational feasibility of the system based on the availability of LGU server infrastructure, computing resources for machine learning model training, internet connectivity, and the capability of administrators to manage the knowledge base via the admin dashboard. Possible risks such as misclassification of queries due to language variations, unsupported inquiries, inaccurate or outdated responses, system downtime, and data privacy concerns will also be considered during the planning stage. Tasks will be structured in a Work Breakdown Structure (WBS), and the development phases and timelines will be presented in a Gantt chart to ensure proper monitoring, organized task distribution, and timely completion of the project.

Project Team Organization

The project team as shown in Figure 7, is organized to ensure effective coordination and clear distribution of responsibilities. The team consists of Four members with the following designated roles.

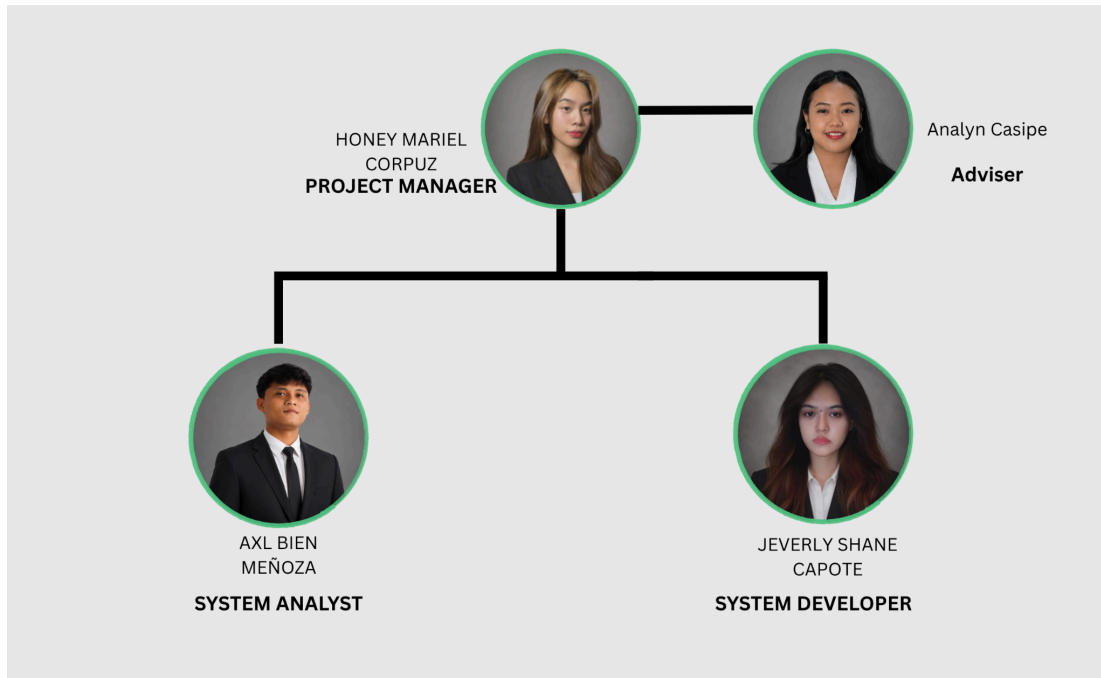


Figure 7. Project Team Organization

The Adviser provides overall guidance and ensures that the project meets academic and technical standards. The Project Manager is responsible for managing the overall workflow, handling technical writing, ensuring all parts of the system are coordinated, finalizing the capstone manuscript, and communicating with stakeholders (LGU Administrators). The System Analyst is primarily responsible for designing the database structure, developing the web interfaces, ensuring a user-friendly conversational UI, and verifying responsive design principles. The System Developer focuses on designing, writing, and testing the core application logic, implementing the machine learning classification models, and integrating the backend with the frontend environment.

Work Breakdown Structure (WBS)

The **Work Breakdown Structure** of the project shown in Figure 8, presents a hierarchical decomposition of the entire system development process into manageable phases and tasks that are specifically aligned with the PADIM framework.

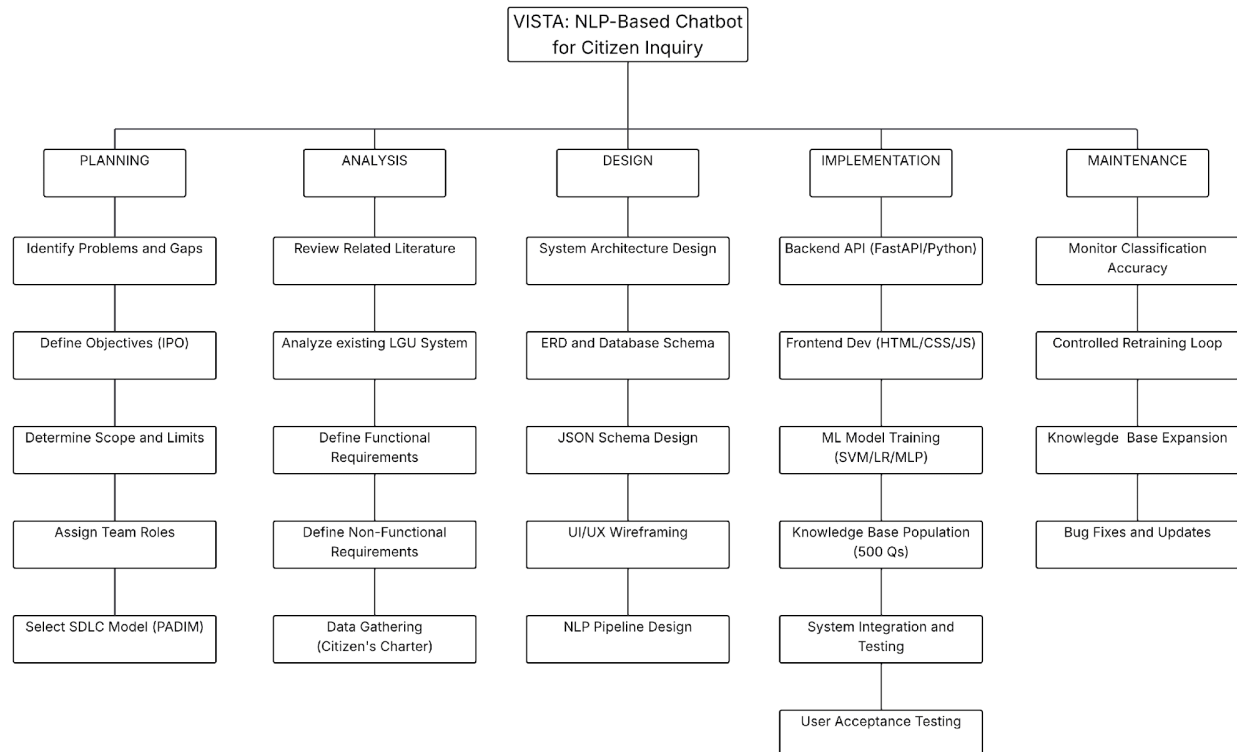


Figure 8. Work Breakdown Structure (WBS)

It begins with the Planning phase where problems are identified, objectives are defined, and the outline and scope of the project is established. It is then followed by Analysis where existing LGU inquiry systems are studied, gaps identified, and functional and non-functional requirements determined. The Design phase covers system architecture development, ERD and database schema creation, UI/UX prototyping, and NLP pipeline design. The Implementation phase encompasses backend API

development, frontend integration, NLP model training and evaluation, and user acceptance testing. Finally, the Maintenance phase focuses on monitoring classification accuracy, executing the controlled retraining loop, expanding the knowledge base, and addressing system bugs and updates.

Gantt Chart

The Gantt chart Shown in Figure 9, presents the timeline and planned schedule for the activities for the development of VISTA, aligning tasks with the PADIM framework.

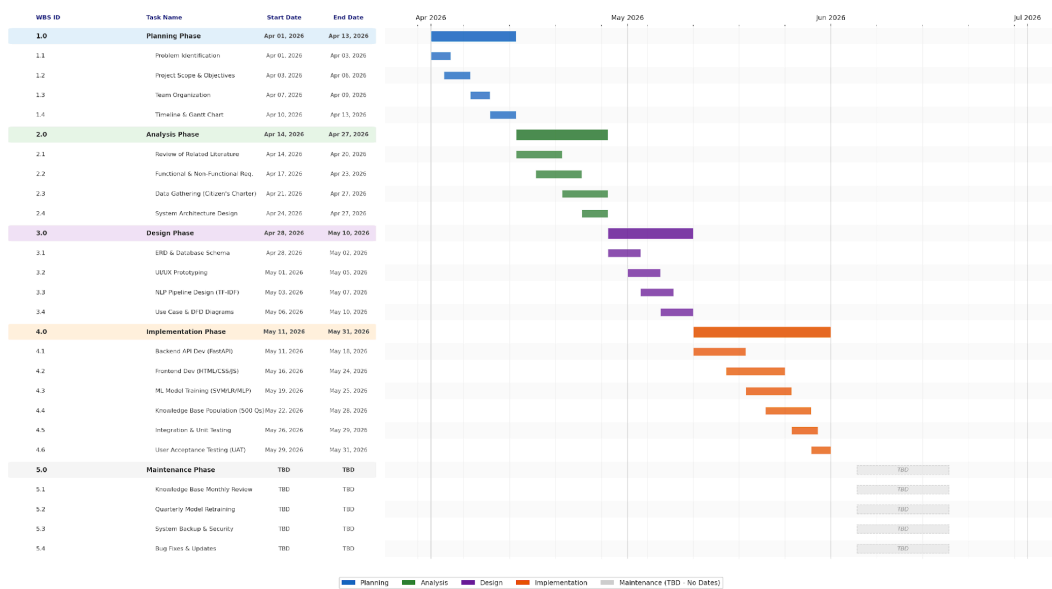


Figure 9. Gantt Chart

The chart showcases the sequence and the duration of phases from Planning to Maintenance, ensuring that all essential activities are completed within the specified timeline. The Planning phase, including Project Initiation, Scoping, and Resource Planning, runs from April to the first week of May 2026. The Analysis and Design phase,

covering Literature Review, System Architecture Design, and ERD/Schema/UI Prototyping, spans from the second week of May to the end of May 2026. The Implementation phase, encompassing Backend Development, Frontend Development, and NLP Model Training, runs from the first week of June through June 1, 2026. The Testing and Evaluation phase and the Maintenance and Documentation phase are scheduled to follow the implementation period; however, specific progress percentages and end dates for these phases are not yet assigned, as they will be determined based on the actual completion of the implementation deliverables. The Gantt chart helps the proponents in monitoring the progress of the study and executing time management effectively.

System Analysis

This section presents the analysis phase of the VISTA system development. It covers the system architecture, functional and non-functional requirements, use case diagram, context flow diagram, and Level 1 data flow diagram.

System Architecture

The system architecture of VISTA Shown in Figure 10, presents a structured framework that illustrates how the different components of the system interact to support real-time citizen inquiry classification, multilingual response retrieval, and administrative knowledge base management. The architecture is composed of three main tiers: Client Tier (Frontend Interface), Application Tier (Backend Server), and Data Tier.

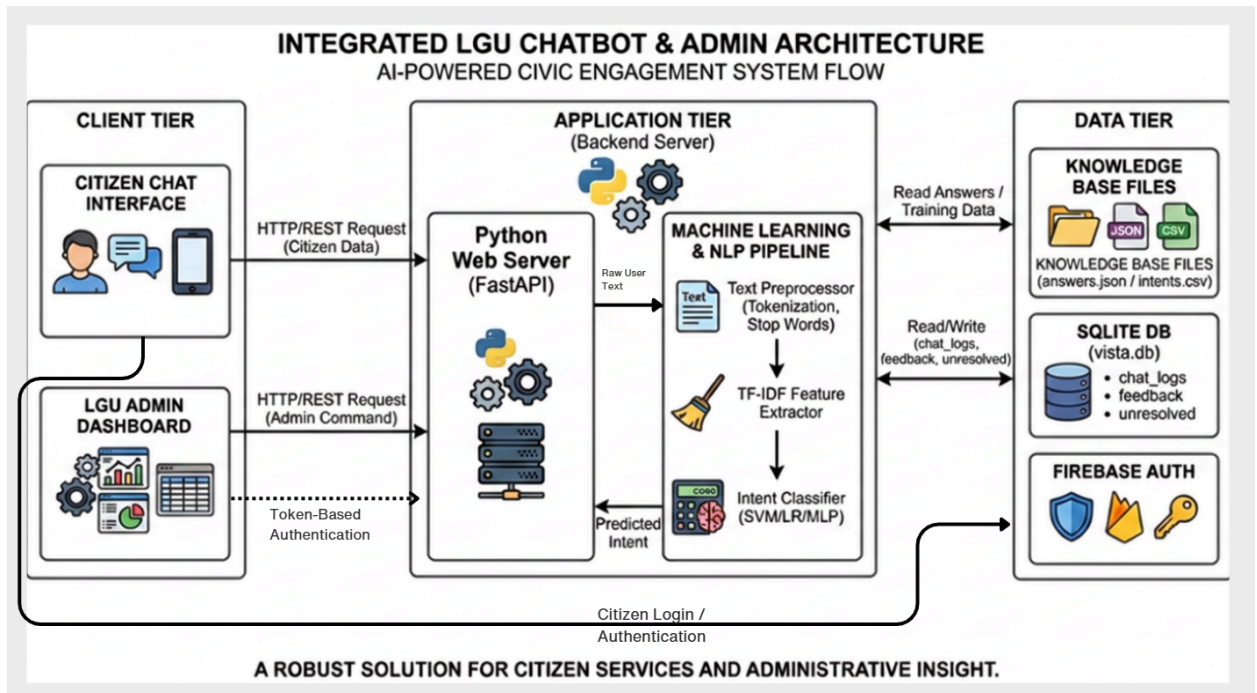


Figure 10. System Architecture

At the Client Tier, the system provides two web-based interfaces: (1) the Citizen Chat Interface, where citizens submit natural language inquiries in English, Filipino/Tagalog, or Cebuano/Bisaya and receive automated responses, utilizing Firebase Authentication for secure citizen login and session management; and (2) the LGU Admin Dashboard, where authorized administrators manage the knowledge base, review unmatched queries, trigger model retraining, and view analytics.

At the Application Tier, the Python-based backend server (FastAPI) receives HTTP/REST requests from both client interfaces. To ensure security, requests originating from the LGU Admin Dashboard are verified using a server-side Token-Based Authentication mechanism. When a citizen submits a query, the server routes the raw text through the Machine Learning and NLP Pipeline, which consists of three sequential stages: (1) the Text Preprocessor performs tokenization, lowercasing,

and stop word removal; (2) the TF-IDF Feature Extractor converts the cleaned text into numerical feature vectors; and (3) the Intent Classifier — using the best-performing model among SVM, Logistic Regression, and Neural Network (MLP) as determined during evaluation — predicts the most probable service intent. The predicted intent is returned to the server, which retrieves the corresponding validated response from the knowledge base.

At the Data Tier, the system utilizes a dual-storage approach to manage information efficiently. A structured file system (utilizing JSON and CSV formats) serves as the Knowledge Base, storing the training data, intent classifications, and validated multilingual answers for fast in-memory retrieval and administrative editing. Simultaneously, an SQLite relational database is employed to store transactional and historical data, specifically the chat logs for descriptive analytics, citizen feedback records, and unmatched query logs for future active learning.

Conceptual Framework

The conceptual framework shown in Figure 11, illustrates how the VISTA system receives citizen text inquiries, processes them through its NLP classification pipeline, and produces automated service responses and administrative analytics. Based on the requirements of Panabo City Hall's frontline offices and citizen users, the framework ensures that the system provides accurate multilingual inquiry handling, knowledge base management, and data-driven insights through a centralized web-based platform.

Input

The input of the system includes raw natural language text submitted by citizens through the chatbot interface. Inquiries may be in English, Filipino/Tagalog, or

Cebuano/Bisaya. The system also receives administrator inputs including knowledge base updates (new intents, edited responses), unmatched query resolutions, and retraining commands through the administrator dashboard. The initial knowledge base content is sourced from the Panabo City Citizen's Charter, covering five frontline offices: City Mayor's Office, City Civil Registrar's Office, City Treasurer's Office, City Assessor's Office, and the City Social Welfare and Development Office.

Process

The process begins when a citizen submits a text query through the chatbot interface. The backend server receives the raw text and passes it through the NLP classification pipeline, which consists of three stages: (1) the Text Preprocessor performs tokenization, lowercasing, and stop word removal to clean the raw input; (2) the TF-IDF Feature Extractor converts the cleaned text into numerical feature vectors; and (3) the Intent Classifier using the best-performing model among SVM, Logistic Regression, and Neural Network (MLP) predicts the most probable service intent. If the classification confidence exceeds the predefined threshold (35%), the system retrieves the corresponding validated response from the knowledge base. If the confidence falls below the threshold, the query is logged as an unmatched query for administrator review. Administrators can also manage the knowledge base, review unmatched queries, and trigger controlled model retraining through the admin dashboard.

Output

The output of the system is a web-based chatbot interface that delivers automated, validated responses to citizen inquiries in real time. The system also provides an administrator dashboard that displays inquiry analytics, including the most

frequently asked service categories, peak inquiry periods, and unmatched query logs. Additional outputs include classification performance reports, exportable chat logs, and a continuously improving knowledge base through the controlled retraining mechanism.

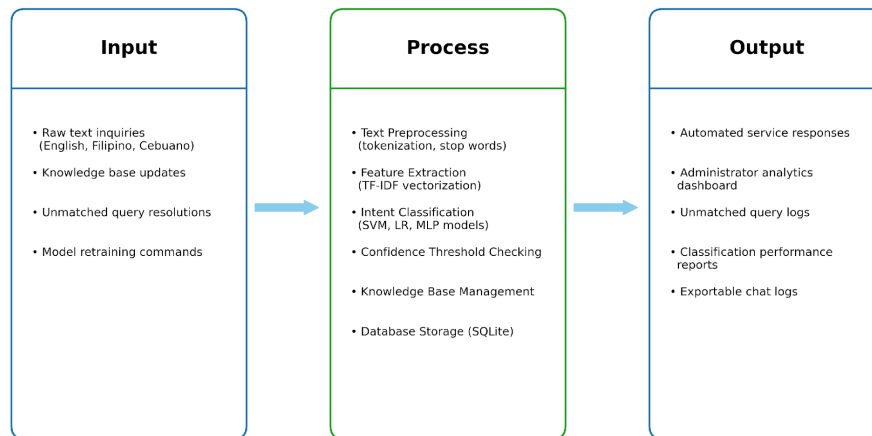


Figure 11. Conceptual Framework

Functional and non-functional requirements

This section discusses the system's expected system functionalities (functional requirements) and expected quality attributes (non-functional requirements).

Functional Requirements:

The system shall provide the following functionalities:

1. The system shall accept natural language text inputs from citizens in English, Filipino/Tagalog, and Cebuano/Bisaya through a web-based chatbot interface.

2. The system shall preprocess raw text input using tokenization, lowercasing, and stop word removal, then extract features using TF-IDF vectorization.
3. The system shall classify the preprocessed input into a predefined service intent category using a trained supervised machine learning model (SVM, Logistic Regression, or Neural Network).
4. The system shall retrieve and display the corresponding validated response from the knowledge base when the classification confidence exceeds the predefined threshold.
5. The system shall log unmatched queries (below confidence threshold) for administrator review and potential incorporation into the training dataset.
6. The system shall provide an administrator dashboard to view inquiry analytics, manage knowledge base intents, review unmatched queries, and trigger controlled model retraining.
7. The system shall allow administrators to securely log in using Firebase Authentication.
8. The system shall display inquiry frequency, top service categories, and peak usage periods through an analytics visualization interface.
9. The system shall support a controlled retraining mechanism where only administrator-validated queries are incorporated into the model.

Non-Functional Requirements:

The following are the system's non-functional requirements:

1. Performance Efficiency. The chatbot shall return responses within 2 seconds of the user query to ensure prompt and satisfactory citizen interaction.
2. Reliability. The backend shall gracefully handle fallback conditions if the intent confidence falls below the 35% threshold, logging unmatched queries and providing a helpful default response.
3. Usability. The web interface shall be responsive and accessible on both desktop and mobile devices, with a conversational UI that is intuitive for citizens with varying technical backgrounds.
4. Security. Administrator passwords shall be securely encrypted and handled via Firebase Authentication, with role-based access control for administrative functions.
5. Compatibility. The system shall be accessible through modern web browsers (Chrome, Firefox, Edge, Safari) on both desktop and mobile platforms.
6. Maintainability. The system shall be designed with modular code to allow for easy debugging, knowledge base updates, model retraining, and future software enhancements.
7. Scalability. The system shall support the addition of new service categories and training data without requiring fundamental architectural changes.

Use Case Diagram

A Use Case Diagram highlights the details of system users' (also known as actors) and interactions with the proposed system. As shown in Figure 12, the primary

actors identified in VISTA are the Citizen/Resident and the LGU Administrator, with supporting roles played by Firebase Authentication and the NLP Classification Engine.

The Citizen/Resident interacts with the system primarily through inquiry and response activities. They submit natural language questions through the chatbot interface, receive classified responses from the knowledge base, provide feedback on response helpfulness, and view related service articles and office directory information. The LGU Administrator is responsible for the overall management and governance of the system. This includes managing the knowledge base (adding, editing, and deleting intent categories and responses), reviewing unmatched queries that fell below the confidence threshold, assigning unmatched queries to appropriate intents, triggering controlled model retraining, viewing inquiry analytics and trend reports, and managing system configurations.

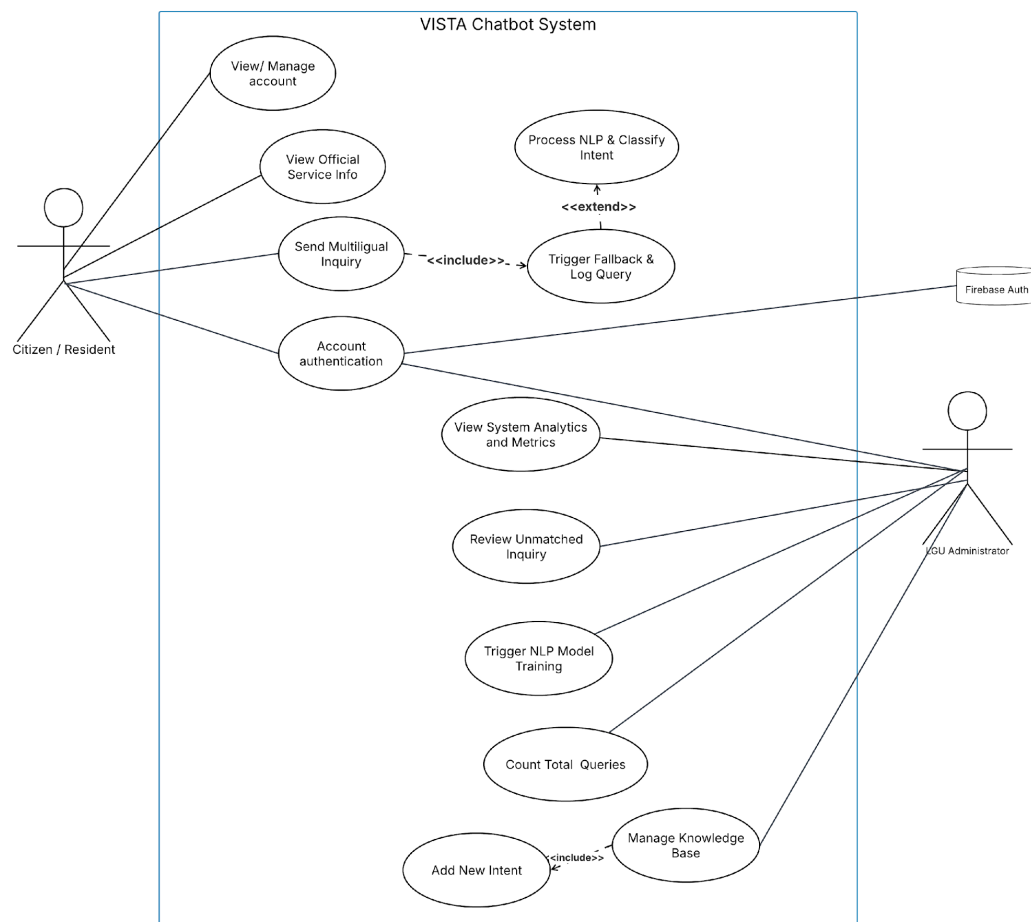


Figure 12. Use Case Diagram

Context Flow Diagram (Level 0 Data Flow Diagram)

The Context Flow Diagram in the current system depicts the flow of business actions (logical). It specifies the business events and the data each event requires and produces. It also shows how the system interacts with external entities (system, organization groups, external data stores, etc.).

In the diagram shown in Figure 13, the three external entities interacting with the VISTA system are the Citizen (User), the LGU admin, and Firebase Auth. The Citizen serves as the primary source of input by submitting user queries and language preferences, user credentials, and feedback. In return, the system provides them with AI responses, related articles, chat history, user account information, and various service details such as LGU services, the office directory, settings details, and help center details. Meanwhile, the LGU admin provides administrative inputs to the system, including admin credentials, directory data management, retrain commands, query review actions, and new intent data. In exchange, the system outputs comprehensive operational data to the admin dashboard, such as system analytics, citizen feedbacks, unmatched queries, system health, total queries, chat logs, as well as current directory and knowledge base details. Finally, Firebase Auth acts as a third-party security entity that receives auth token requests from the system and returns the corresponding auth tokens and user profiles to facilitate secure login for both citizens and administrators.

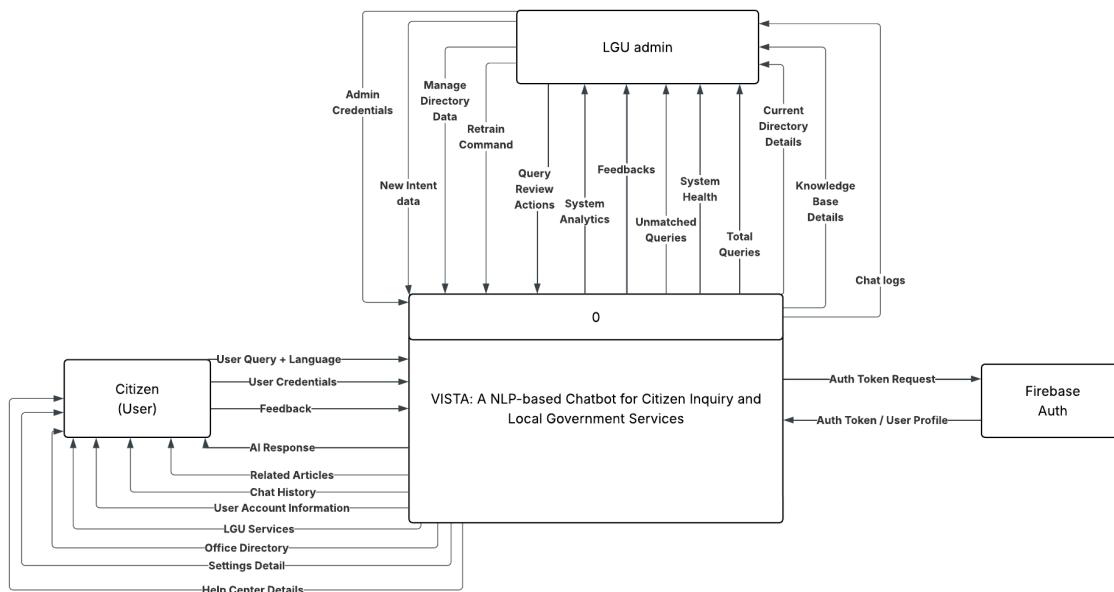


Figure 13. Context Flow **Diagram** (Level 0 DFD)

Level 1 Data Flow Diagram

The Level 1 Data Flow Diagram shown in Figure 14, decomposes the VISTA system into its constituent sub-processes, providing a more detailed view of how data flows between system modules. The external entities and data flows established in the Context Flow Diagram (Level 0 DFD) are retained and expanded upon in this diagram.

In this diagram, the main VISTA system (from Level 0) is decomposed into eight key operational modules: (1.0) User Authentication, (2.0) Chat Query Processing, (3.0) Feedback Management, (4.0) Content & Service Display, (5.0) Knowledge Base Management, (6.0) Model Retraining, (7.0) Analytics & Reporting, and (8.0) Directory Management. Furthermore, while the system handles various types of data, the data

stores in this diagram are consolidated into two primary data stores: D1 (Knowledge Base), which contains intents, training data, and the service directory; and D2 (System Logs), which contains chat logs, unmatched queries, and analytics data. This consolidation accurately reflects the actual system design, where a single SQLite database efficiently manages all persistent data through relational tables.

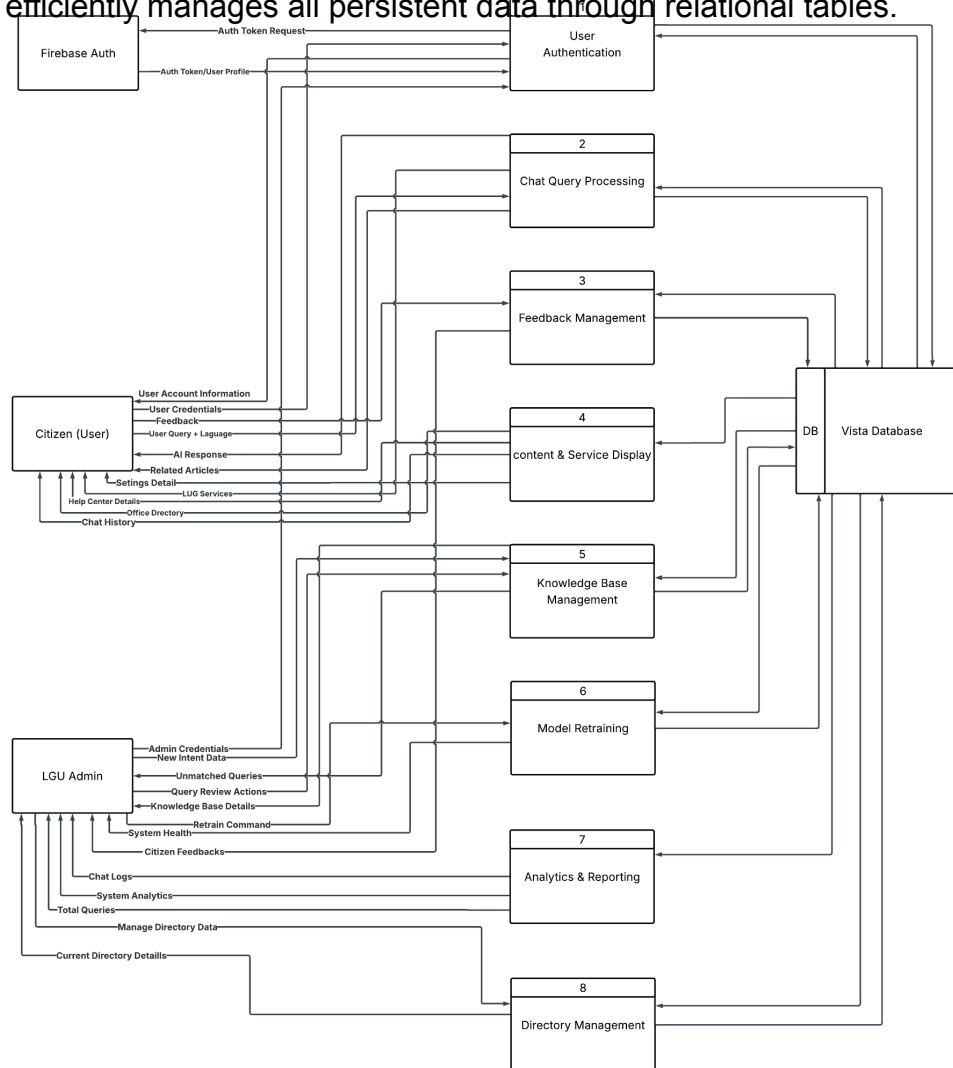


Figure 14. Level 1 Data Flow Diagram

System Design

This section presents the design phase of the VISTA system development, detailing the database structure through the Entity Relationship Diagram, JSON schema design, data dictionary, and the technologies, concepts, and theories used in developing the system.

Entity Relationship Diagram (ERD)

The Entity Relationship Diagram (ERD) shown in Figure 15, defines the structure of the database for VISTA, showing how data entities are organized and related to one another.

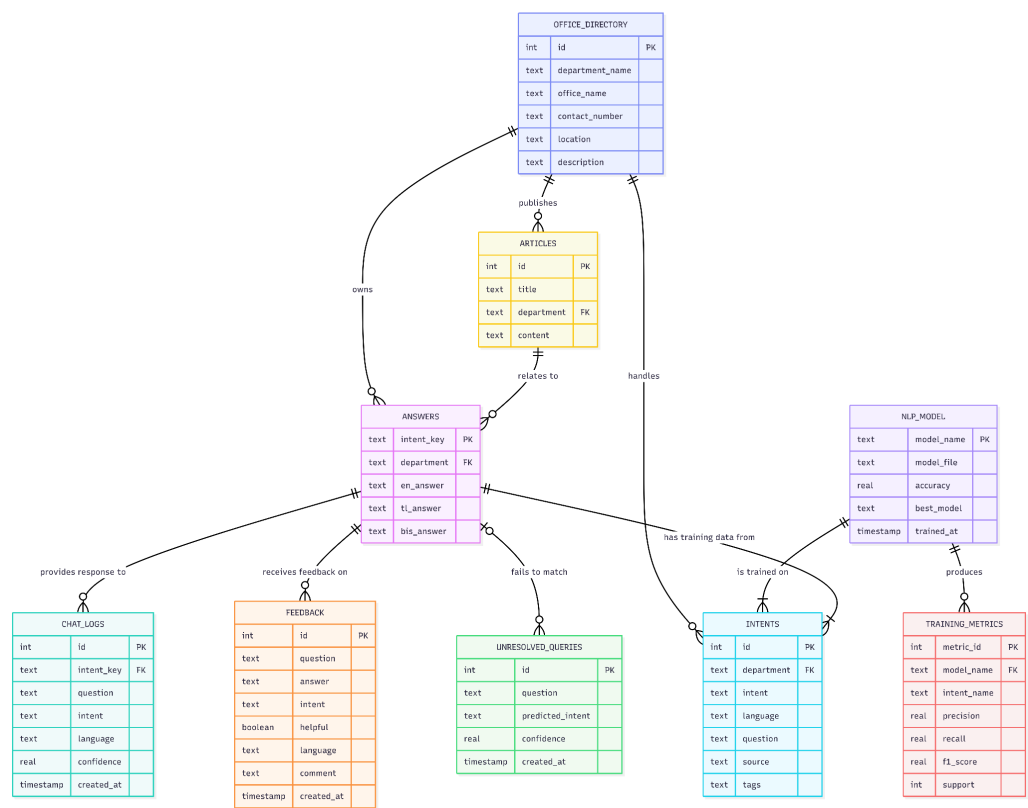


Figure 15. Entity Relationship Diagram (ERD)

The Entity Relationship Diagram illustrates the relational database schema supporting the VISTA system. The ANSWERS entity serves as a core component, storing the validated response texts in multiple languages (English, Tagalog, Bisaya) sourced from the Citizen's Charter. The INTENTS entity acts as the training data repository, storing individual user questions, their assigned intent tags, languages, and source documents, which are used to train the machine learning models (NLP_MODEL). The performance evaluations for these models are tracked in the TRAINING_METRICS entity. Citizen inquiries are recorded in the CHAT_LOGS entity, which logs the user's original question, detected language, predicted intent, the model's confidence score, and a timestamp. User feedback regarding the chatbot's responses is collected in the FEEDBACK entity. Queries that fail to match an intent with high confidence are stored in the UNRESOLVED_QUERIES entity, pending administrator review. To support broader information retrieval, the ARTICLES entity stores supplementary knowledge base content. Finally, the OFFICE_DIRECTORY entity manages organizational data, including department names, office names, contact numbers, locations, and descriptions.

JSON Schema Diagram

Although VISTA uses an SQL-based database (SQLite) as its primary data store, a JSON schema design shown in Figure 16, is also utilized to represent key data components such as intent definitions, training phrases, and API request/response structures. The use of JSON Schema supports interoperability between different system components and external services.

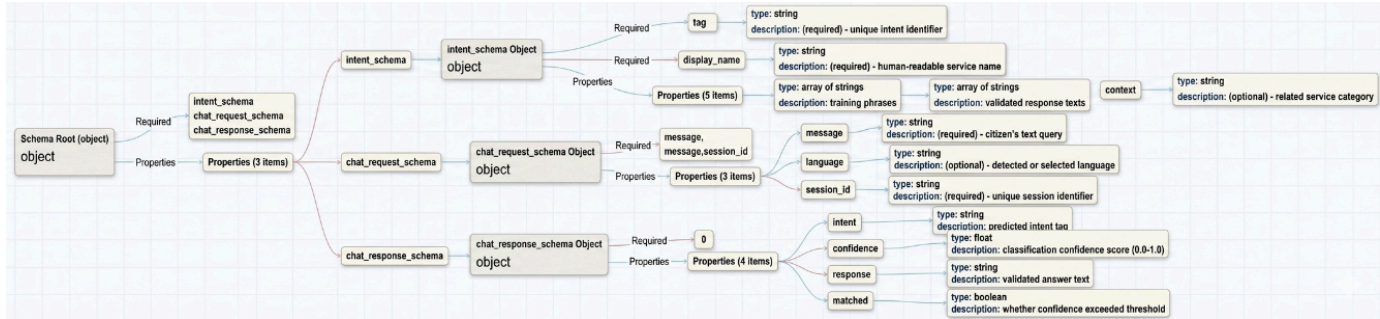


Figure 16. JSON Schema Design

The schema defines validation rules such as required fields, data types, and formatting standards. These rules ensure that incoming data from citizen queries and administrative inputs are properly validated before being processed by the classification pipeline, reducing errors and enhancing overall system reliability.

Data Dictionary

The data dictionary of the system displayed in this section presents a detailed description of all data elements used in VISTA, serving as a reference for the system's database structure and data organization.

Table 2. List of Database Tables

Table Name	Description
OFFICE_DIRECTORY	Manages organizational data including department names, locations, contacts, and descriptions.
ARTICLES	Stores supplementary knowledge base content and announcements linked to specific departments.
ANSWERS	Core repository for official, validated LGU service responses in English, Tagalog, and Bisaya.

INTENTS	Dataset of labeled sample questions and intent tags used for training the machine learning model.
NLP_MODEL	Stores metadata about trained classification models, including accuracy and file paths.
TRAINING_METRICS	Records performance evaluation metrics (precision, recall, f1-score) for the models across intents.
CHAT_LOGS	Historical records of all citizen-chatbot interactions, tracking predicted intents and confidence scores.
FEEDBACK	Collects citizen ratings and comments on the helpfulness of the chatbot's provided answers.
UNRESOLVED_QUERIES	Records citizen questions that fell below the confidence threshold, pending administrator review.

Table 2.1. OFFICE_DIRECTORY Table

Field Name	Data Type	Description
id	INTEGER (PK)	Auto-incremented unique identifier for the office.
department_name	TEXT	Name of the department.
office_name	TEXT	Name of the specific office or division.
contact_number	TEXT	Contact details for the office.
location	TEXT	Physical location or building of the office.

description	TEXT	Brief description of the office's functions and services.
-------------	------	---

Table 2.2. ARTICLES Table

Field Name	Data Type	Description
id	INTEGER (PK)	Auto-incremented unique identifier for the article.
title	TEXT	Title of the article or announcement.
department	TEXT (FK)	Foreign key linking to the associated department.
content	TEXT	Full text content of the article.

Table 2.3. ANSWERS Table

Field Name	Data Type	Description
intent_key	TEXT (PK)	Unique string identifier (intent tag) for the service category.
department	TEXT (FK)	Foreign key linking to the responsible department.
en_answer	TEXT	Validated response text in English.
tl_answer	TEXT	Validated response text in Tagalog (Filipino).
bis_answer	TEXT	Validated response text in Bisaya (Cebuano).

Table 2.4. INTENTS Table

Field Name	Data Type	Description
------------	-----------	-------------

id	INTEGER (PK)	Auto-incremented unique identifier for the training sample.
department	TEXT (FK)	Foreign key linking to the associated department.
intent	TEXT	The target intent classification tag.
language	TEXT	Language of the question (e.g., en, tl, bis).
question	TEXT	Sample citizen query used for training the NLP model.
source	TEXT	Origin of the training data (e.g., citizen_charter, user_logs).
tags	TEXT	Keywords associated with the intent.

Table 2.5. NLP_MODEL Table

Field Name	Data Type	Description
model_name	TEXT (PK)	Unique identifier/name for the trained model.
model_file	TEXT	File path where the serialized model is stored.
accuracy	REAL	Overall classification accuracy score of the model.
best_model	TEXT	Boolean or string flag indicating if it is the current production model.
trained_at	TIMESTAMP	Date and time when the model was trained.

Table 2.6. TRAINING_METRICS Table

Field Name	Data Type	Description
------------	-----------	-------------

metric_id	INTEGER (PK)	Auto-incremented unique identifier for the metric record.
model_name	TEXT (FK)	Foreign key linking to the NLP_MODEL table.
intent_name	TEXT	Specific intent category being evaluated.
precision	REAL	Precision score for the intent classification.
recall	REAL	Recall score for the intent classification.
f1_score	REAL	F1-score for the intent classification.
support	INTEGER	Number of true instances for the intent in the test set.

Table 2.7. CHAT_LOGS Table

Field Name	Data Type	Description
id	INTEGER (PK)	Auto-incremented unique identifier for the chat log entry.
intent_key	TEXT (FK)	Foreign key linking to the matched intent in ANSWERS.
question	TEXT	Original text query submitted by the citizen.
intent	TEXT	The intent predicted by the NLP classifier.
language	TEXT	Language detected from the user's input.
confidence	REAL	Classification confidence score of the prediction.
created_at	TIMESTAMP	Timestamp of the interaction.

Table 2.8. FEEDBACK Table

Field Name	Data Type	Description
id	INTEGER (PK)	Auto-incremented unique identifier for the feedback entry.
question	TEXT	Original query that triggered the feedback.
answer	TEXT	The response provided by the chatbot.
intent	TEXT	The predicted intent associated with the interaction.
helpful	BOOLEAN	True if the citizen found the answer helpful, False otherwise.
language	TEXT	Language of the interaction.
comment	TEXT	Optional textual feedback from the citizen.
created_at	TIMESTAMP	Timestamp when the feedback was submitted.

Table 2.9. UNRESOLVED_QUERIES Table

Field Name	Data Type	Description
id	INTEGER (PK)	Auto-incremented unique identifier for the unresolved query.
question	TEXT	Original text query submitted by the citizen that failed to match.
predicted_intent	TEXT	The highest scoring intent, though below threshold.

confidence	REAL	The below-threshold confidence score.
created_at	TIMESTAMP	Timestamp when the unresolved query was logged.

Technologies, Concepts, and Theories

This section deeply discusses the model/ algorithm/ emerging technologies shown in the system architecture. It includes the discussion of the following but is not limited to, depending on the study's objectives and the process involved in the model.

Data Collection

Data collection is performed through manual extraction and digitization of service information from the official Panabo City Citizen's Charter (CMO 2026 edition), supplemented by historical citizen inquiries sourced from the LGU's public Facebook Messenger page. The Citizen's Charter provides structured information including service names, requirements, procedures, fees, and processing times for all frontline offices. The dataset used in this study is a custom-built dataset constructed entirely by the developers from these primary LGU sources; no pre-existing or downloaded dataset was used, as no publicly available training corpus exists for Philippine LGU citizen inquiries in English, Filipino/Tagalog, and Cebuano/Bisaya.

Data Pre-processing

Before classification, the collected text data undergoes pre-processing to improve its quality and reliability [28]. This includes tokenization (splitting sentences into individual words), lowercasing (converting all text to lowercase for uniformity), and stop

word removal (filtering common conversational words such as "the," "is," "and," "a" that do not contribute to intent meaning). Pre-processing is performed using Python libraries including NLTK and Scikit-learn's built-in text utilities. This ensures that the input data is clean and suitable for feature extraction, reducing the likelihood of misclassification caused by noise in the text.

Feature Extraction

Feature extraction involves transforming the cleaned text into numerical format suitable for machine learning algorithms. VISTA utilizes Term Frequency-Inverse Document Frequency (TF-IDF) vectorization, which evaluates how relevant a word is to a document within a collection of documents [17]. TF-IDF assigns higher weights to distinctive, domain-specific keywords (e.g., "permit," "clearance," "cedula," "business") and lower weights to common words that appear across many documents.

Classification

The system evaluates three supervised machine learning algorithms to determine the optimal classifier for VISTA's intent classification pipeline. The purpose of these algorithms is intent classification — when a citizen types a message, the algorithm analyzes the processed text, determines what the user's intent or goal is, and matches it to the correct FAQ category in the knowledge base so the chatbot can reply with an accurate, validated response.

Support Vector Machine (SVM) [18] creates an optimal hyperplane to separate different intent categories in high-dimensional TF-IDF feature space. SVM is particularly

effective for text classification because it handles the high dimensionality of text features well and provides robust decision boundaries even with limited training data.

Logistic Regression [19] is a probabilistic model that estimates the probability of a query belonging to each possible intent category. It provides interpretable confidence scores and serves as a reliable baseline classifier for multiclass text problems.

Neural Network — Multilayer Perceptron (MLP) [20] is a feedforward artificial neural network consisting of an input layer, one or more hidden layers, and an output layer. MLP classifiers learn non-linear patterns and complex feature interactions through backpropagation, often achieving higher accuracy than linear models when sufficient training data is available.

All three algorithms are implemented and evaluated using the Scikit-learn machine learning library [21]. The model achieving the highest combined accuracy, precision, recall, and F1-score across all intent categories is selected for production deployment.

This part also includes a list of the technologies required to develop the system. A definition of the technology, a snapshot of the Interface as the proponents used it, and a discussion of how the technology was utilized are included in the manuscript.

Python Programming Language. As shown in Figure 17, Python serves as the primary backend programming language for VISTA. It was selected because it provides the most robust and mature ecosystem of Natural Language Processing and machine learning libraries available, including NLTK for text preprocessing and Scikit-learn for model training and evaluation. Python's syntax is clean and readable, which accelerates

development and debugging. Furthermore, the FastAPI web framework, built on Python, handles all HTTP/REST API endpoints for both the chatbot and administrator interfaces, providing high-performance asynchronous request handling suitable for real-time chatbot interactions [29].

```
backend > app.py > ...
1  import os
2  import json
3  import joblib
4  import pandas as pd
5  from datetime import datetime
6  from fastapi import FastAPI, Depends, HTTPException, Header
7  from fastapi.middleware.cors import CORSMiddleware
8  from pydantic import BaseModel
9
10 from utils.preprocess import clean_text
11 from utils.language_detector import detect_language
12 from utils.article_search import search_related_articles
13 from train import train_and_compare
14 from database import get_db
15 from dotenv import load_dotenv
16
17 load_dotenv()
18 ADMIN_TOKEN_SECRET = os.getenv("ADMIN_TOKEN", "vista_admin_2026")
19
20 app = FastAPI(title="VISTA API")
21
22 def verify_admin_token(x_admin_token: str = Header(None)):
23     if x_admin_token != ADMIN_TOKEN_SECRET:
24         raise HTTPException(status_code=401, detail="Unauthorized")
25
26
27 app.add_middleware(
28     CORSMiddleware,
29     allow_origins=["*"],
30     allow_credentials=True,
31     allow_methods=["*"],
32     allow_headers=["*"],
33 )
34
35 model = joblib.load("models/best_model.pkl")
36 label_encoder = joblib.load("models/label_encoder.pkl")
37
```

Figure 17. Python Source Code in VS Code

Scikit-learn Machine Learning Library. Scikit-learn [21] shown in Figure 18, provides the implementation of all three classification algorithms (SVM, Logistic Regression, Neural Network/MLP), TF-IDF vectorization, model evaluation metrics, and the complete machine learning pipeline infrastructure. It was chosen because it is the industry standard open-source library for traditional machine learning in Python, offering

well-documented and computationally efficient implementations that are suitable for deployment on resource-constrained LGU servers without requiring GPU hardware [21].

```
from sklearn.model_selection import train_test_split
from sklearn.pipeline import Pipeline
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score, classification_report

from sklearn.linear_model import LogisticRegression
from sklearn.svm import LinearSVC
from sklearn.neural_network import MLPClassifier

from utils.preprocess import clean_text

DATA_PATH = "data/intents.csv"
MODEL_DIR = "models/"

def load_data():
    df = pd.read_csv(DATA_PATH)
    df["question"] = df["question"].astype(str).apply(clean_text)
    return df

def train_and_compare():
    df = load_data()

    X = df["question"]
    y = df["intent"]

    label_encoder = LabelEncoder()
    y_encoded = label_encoder.fit_transform(y)

    X_train, X_test, y_train, y_test = train_test_split(
        X, y_encoded, test_size=0.2, random_state=42
    )

    models = {
        "logistic_regression": LogisticRegression(max_iter=1000, random_state=42),
        "svm": LinearSVC(),
        "neural_network": MLPClassifier(hidden_layer_sizes=(100), max_iter=1000, random_state=42)
    }

    results = {}
    best_name = None
    best_score = 0
    best_pipeline = None

    for name, clf in models.items():
        pipeline = Pipeline([
            ("tfidf", TfidfVectorizer(ngram_range=(1, 2))),
            ("clf", clf)
        ])

        pipeline.fit(X_train, y_train)
        y_pred = pipeline.predict(X_test)
        acc = accuracy_score(y_test, y_pred)
```

Figure 18. Scikit-learn Implementation in Source Code

Natural Language Toolkit (NLTK). NLTK provides essential text preprocessing utilities including tokenization, stop word lists, and stemming functions used to clean and normalize raw citizen text input as shown in Figure 19. NLTK was selected because it is one of the most widely used and well-documented NLP libraries for Python, offering

comprehensive support for text processing tasks that are foundational to the system's classification pipeline [30].

```

1 import re
2
3 STOP_WORDS = set([
4     # English
5     "a", "an", "the", "is", "are", "to", "from", "of", "and", "in", "on", "with", "how", "what", "where", "when", "why", "can", "it", "you", "do", "does", "my", "we",
6     # Tagalog
7     "ang", "nga", "sa", "na", "ng", "ay", "ba", "pa", "po", "opo", "ko", "mo", "niya", "kami", "tayo", "nila", "kung", "para", "ano", "paano", "saan", "kailan", "bakit",
8     # Bisaya
9     "ang", "nga", "sa", "nga", "ug", "kay", "ba", "pa", "ko", "mo", "nimo", "nako", "niya", "kami", "kita", "nila", "kung", "para", "unsa", "unsaon", "asa", "kanus-a", "ngano", "ka", "og"
10 ])
11
12 def clean_text(text: str) -> str:
13     # Lowercase and strip whitespace
14     text = text.lower().strip()
15     # Remove special characters but keep letters and numbers
16     text = re.sub(r"[^a-zA-Z0-9\u00a5\u00d7\u0021\u0022\u0023\u0024\u0025\u0026\u0027\u0028\u0029\u002a\u002b\u002c\u002d\u002e\u002f\u0030\u0031\u0032\u0033\u0034\u0035\u0036\u0037\u0038\u0039\u0040\u0041\u0042\u0043\u0044\u0045\u0046\u0047\u0048\u0049\u004a\u004b\u004c\u004d\u004e\u004f\u0050\u0051\u0052\u0053\u0054\u0055\u0056\u0057\u0058\u0059\u005a\u005b\u005c\u005d\u005e\u005f\u0060\u0061\u0062\u0063\u0064\u0065\u0066\u0067\u0068\u0069\u006a\u006b\u006c\u006d\u006e\u006f\u0070\u0071\u0072\u0073\u0074\u0075\u0076\u0077\u0078\u0079\u007a\u007b\u007c\u007d\u007e\u007f\u0080\u0081\u0082\u0083\u0084\u0085\u0086\u0087\u0088\u0089\u008a\u008b\u008c\u008d\u008e\u008f\u0090\u0091\u0092\u0093\u0094\u0095\u0096\u0097\u0098\u0099\u009a\u009b\u009c\u009d\u009e\u009f\u00a0\u00a1\u00a2\u00a3\u00a4\u00a5\u00a6\u00a7\u00a8\u00a9\u00aa\u00ab\u00ac\u00ad\u00ae\u00af\u00b0\u00b1\u00b2\u00b3\u00b4\u00b5\u00b6\u00b7\u00b8\u00b9\u00ba\u00bb\u00bc\u00bd\u00be\u00bf\u00c0\u00c1\u00c2\u00c3\u00c4\u00c5\u00c6\u00c7\u00c8\u00c9\u00ca\u00cb\u00cc\u00cd\u00ce\u00cf\u00d0\u00d1\u00d2\u00d3\u00d4\u00d5\u00d6\u00d7\u00d8\u00d9\u00da\u00db\u00dc\u00dd\u00de\u00df\u00e0\u00e1\u00e2\u00e3\u00e4\u00e5\u00e6\u00e7\u00e8\u00e9\u00ea\u00eb\u00ec\u00ed\u00í\u00f0\u00f1\u00f2\u00f3\u00f4\u00f5\u00f6\u00f7\u00f8\u00f9\u00fa\u00fb\u00fc\u00fd\u00fe\u00ff\u0100\u0101\u0102\u0103\u0104\u0105\u0106\u0107\u0108\u0109\u010a\u010b\u010c\u010d\u010e\u010f\u0110\u0111\u0112\u0113\u0114\u0115\u0116\u0117\u0118\u0119\u011a\u011b\u011c\u011d\u011e\u011f\u0120\u0121\u0122\u0123\u0124\u0125\u0126\u0127\u0128\u0129\u012a\u012b\u012c\u012d\u012e\u012f\u0130\u0131\u0132\u0133\u0134\u0135\u0136\u0137\u0138\u0139\u013a\u013b\u013c\u013d\u013e\u013f\u0140\u0141\u0142\u0143\u0144\u0145\u0146\u0147\u0148\u0149\u014a\u014b\u014c\u014d\u014e\u014f\u0150\u0151\u0152\u0153\u0154\u0155\u0156\u0157\u0158\u0159\u015a\u015b\u015c\u015d\u015e\u015f\u0160\u0161\u0162\u0163\u0164\u0165\u0166\u0167\u0168\u0169\u016a\u016b\u016c\u016d\u016e\u016f\u0170\u0171\u0172\u0173\u0174\u0175\u0176\u0177\u0178\u0179\u017a\u017b\u017c\u017d\u017e\u017f\u0180\u0181\u0182\u0183\u0184\u0185\u0186\u0187\u0188\u0189\u018a\u018b\u018c\u018d\u018e\u018f\u0190\u0191\u0192\u0193\u0194\u0195\u0196\u0197\u0198\u0199\u019a\u019b\u019c\u019d\u019e\u019f\u01a0\u01a1\u01a2\u01a3\u01a4\u01a5\u01a6\u01a7\u01a8\u01a9\u01aa\u01ab\u01ac\u01ad\u01ae\u01af\u01b0\u01b1\u01b2\u01b3\u01b4\u01b5\u01b6\u01b7\u01b8\u01b9\u01ba\u01bb\u01bc\u01bd\u01be\u01bf\u01c0\u01c1\u01c2\u01c3\u01c4\u01c5\u01c6\u01c7\u01c8\u01c9\u01ca\u01cb\u01cc\u01cd\u01ce\u01cf\u01d0\u01d1\u01d2\u01d3\u01d4\u01d5\u01d6\u01d7\u01d8\u01d9\u01da\u01db\u01dc\u01dd\u01de\u01df\u01e0\u01e1\u01e2\u01e3\u01e4\u01e5\u01e6\u01e7\u01e8\u01e9\u01ea\u01eb\u01ec\u01ed\u01ee\u01ef\u01f0\u01f1\u01f2\u01f3\u01f4\u01f5\u01f6\u01f7\u01f8\u01f9\u01fa\u01fb\u01fc\u01fd\u01fe\u01ff\u0200\u0201\u0202\u0203\u0204\u0205\u0206\u0207\u0208\u0209\u020a\u020b\u020c\u020d\u020e\u020f\u0210\u0211\u0212\u0213\u0214\u0215\u0216\u0217\u0218\u0219\u021a\u021b\u021c\u021d\u021e\u021f\u0220\u0221\u0222\u0223\u0224\u0225\u0226\u0227\u0228\u0229\u022a\u022b\u022c\u022d\u022e\u022f\u0230\u0231\u0232\u0233\u0234\u0235\u0236\u0237\u0238\u0239\u023a\u023b\u023c\u023d\u023e\u023f\u0240\u0241\u0242\u0243\u0244\u0245\u0246\u0247\u0248\u0249\u024a\u024b\u024c\u024d\u024e\u024f\u0250\u0251\u0252\u0253\u0254\u0255\u0256\u0257\u0258\u0259\u025a\u025b\u025c\u025d\u025e\u025f\u0260\u0261\u0262\u0263\u0264\u0265\u0266\u0267\u0268\u0269\u026a\u026b\u026c\u026d\u026e\u026f\u0270\u0271\u0272\u0273\u0274\u0275\u0276\u0277\u0278\u0279\u027a\u027b\u027c\u027d\u027e\u027f\u0280\u0281\u0282\u0283\u0284\u0285\u0286\u0287\u0288\u0289\u028a\u028b\u028c\u028d\u028e\u028f\u0290\u0291\u0292\u0293\u0294\u0295\u0296\u0297\u0298\u0299\u029a\u029b\u029c\u029d\u029e\u029f\u02a0\u02a1\u02a2\u02a3\u02a4\u02a5\u02a6\u02a7\u02a8\u02a9\u02aa\u02ab\u02ac\u02ad\u02ae\u02af\u02b0\u02b1\u02b2\u02b3\u02b4\u02b5\u02b6\u02b7\u02b8\u02b9\u02ba\u02bb\u02bc\u02bd\u02be\u02bf\u02c0\u02c1\u02c2\u02c3\u02c4\u02c5\u02c6\u02c7\u02c8\u02c9\u02ca\u02cb\u02cc\u02cd\u02ce\u02cf\u02d0\u02d1\u02d2\u02d3\u02d4\u02d5\u02d6\u02d7\u02d8\u02d9\u02da\u02db\u02dc\u02dd\u02de\u02df\u02e0\u02e1\u02e2\u02e3\u02e4\u02e5\u02e6\u02e7\u02e8\u02e9\u02ea\u02eb\u02ec\u02ed\u02ee\u02ef\u02f0\u02f1\u02f2\u02f3\u02f4\u02f5\u02f6\u02f7\u02f8\u02f9\u02fa\u02fb\u02fc\u02fd\u02fe\u02ff\u0300\u0301\u0302\u0303\u0304\u0305\u0306\u0307\u0308\u0309\u030a\u030b\u030c\u030d\u030e\u030f\u0310\u0311\u0312\u0313\u0314\u0315\u0316\u0317\u0318\u0319\u031a\u031b\u031c\u031d\u031e\u031f\u0320\u0321\u0322\u0323\u0324\u0325\u0326\u0327\u0
```

Figure 19. NLTK Implementation in Source Code

SQLite Database. SQLite serves as the lightweight, serverless relational database management system for storing intents, training data, chat logs, unmatched queries, and administrator credentials that is shown in Figure 20. It was chosen because its self-contained, zero-configuration nature makes it ideal for local deployment on LGU servers without requiring a separate database server or complex installation procedures, thereby reducing the technical burden on LGU IT staff [31].

```

10 def init_db():
11     """Initializes the SQLite database with the necessary tables."""
12     with get_db() as conn:
13         cursor = conn.cursor()
14
15         # Chat Logs Table
16         cursor.execute('''
17             CREATE TABLE IF NOT EXISTS chat_logs (
18                 id INTEGER PRIMARY KEY AUTOINCREMENT,
19                 question TEXT NOT NULL,
20                 intent TEXT,
21                 language TEXT,
22                 confidence REAL,
23                 created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
24             )
25         ''')
26
27         # Feedback Table
28         cursor.execute('''
29             CREATE TABLE IF NOT EXISTS feedback (
30                 id INTEGER PRIMARY KEY AUTOINCREMENT,
31                 question TEXT NOT NULL,
32                 answer TEXT NOT NULL,
33                 intent TEXT,
34                 helpful BOOLEAN,
35                 language TEXT,
36                 comment TEXT,
37                 created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
38             )
39         ''')
40

```

Figure 20. SQLite Database Schema

Firestore Authentication. Firestore Authentication provides secure citizen login and session management, allowing residents to securely save their chat history across devices using Google, Email, or Phone OTP. As shown in Figure 21, the client-side application initializes the core Firestore configuration parameters to instantiate the authentication instance. It was selected because it offers a reliable, free-tier authentication service that handles password hashing, session tokens, and secure credential storage without requiring the developers to build a custom authentication system from scratch [32].

```
const firebaseConfig = {
  apiKey: "XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX",
  authDomain: "XXXXXXXXXXXXXXXXXXXX.firebaseio.com",
  projectId: "XXXXXXXXXXXXXXXXXXXX",
  storageBucket: "XXXXXXXXXXXXXXXXXXXX.firebaseio.com",
  messagingSenderId: "XXXXXXXXXXXXXXXXXXXX",
  appId: "XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX",
  measurementId: "G-XXXXXXXXXXXXXXXXXXXX"
};

firebase.initializeApp(firebaseConfig);
const auth = firebase.auth();
```

Figure 21. Integration of Firebase Auth in the VISTA Client Application

HTML/CSS/JavaScript Frontend. The citizen chatbot interface and administrator dashboard are built using standard web technologies (HTML5, CSS3, JavaScript) to ensure cross-browser compatibility and responsive design on both desktop and mobile devices as shown in Figure 22. These technologies were chosen because they are universally supported by all modern web browsers without requiring additional plugins or frameworks, ensuring maximum accessibility for citizens regardless of their device or browser choice [33].

```
1 <!DOCTYPE html>
2 <html lang="en">
3
4 <head>
5   <meta charset="UTF-8" />
6   <meta name="viewport" content="width=device-width, initial-scale=1.0" />
7   <meta name="description"
8     content="VISTA is an AI-powered trilingual chatbot for IGD citizen inquiries. Ask about permits, certificates, taxes, and government services in English, Tagalog, or Bisaya." />
9   <title>VISTA - IGD Citizen Assistant</title>
10  <link rel="icon"
11    href="data:image/png;base64=http://www.igd.org/2000/svg" class="img" />
12  <type="image/svg+xml" />
13  <link rel="preconnect" href="https://fonts.googleapis.com" />
14  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin />
15  <link href="https://fonts.googleapis.com/css2?family=Inter:wght@200;300;400;500;600;700;800&display=swap" rel="stylesheet" />
16  <link rel="stylesheet" href="style.css" />
17 </head>
18
19 <body class="light-theme">
20   <div class="app" id="app">
```

Figure 22. Frontend Source Code

System Development and Testing

This section presents the results of the machine learning model evaluation conducted during the development of VISTA. The evaluation compares the performance of the three candidate classification algorithms, Support Vector Machine (SVM), Logistic Regression, and Neural Network (Multilayer Perceptron), using the custom-built intent classification dataset derived from the Panabo City Citizen's Charter.

To determine the best-performing algorithm for VISTA's intent classification pipeline, the three supervised machine learning models, Support Vector Machine (SVM), Logistic Regression, and Neural Network (Multilayer Perceptron), were trained and evaluated using the custom-built dataset in Google Colab.

The evaluation uses four standard machine learning metrics computed from the Confusion Matrix [34]. Accuracy is the ratio of correctly predicted intents to the total number of predictions, calculated as: $\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$, where TP is True Positive, TN is True Negative, FP is False Positive, and FN is False Negative. Precision is the ratio of true positive predictions to all positive predictions for each intent category, calculated as: $\text{Precision} = TP / (TP + FP)$, measuring the exactness of the classifier. Recall is the ratio of true positive predictions to all actual instances of each intent category, calculated as: $\text{Recall} = TP / (TP + FN)$, measuring the completeness of the classifier. F1-Score is the harmonic mean of Precision and Recall, calculated as: $F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$, providing a balanced measure of classification performance [35].

The raw performance breakdown for each intent category across the tested models is shown in Figure 23, which displays the generated classification report results exported directly from the Google Colab environment.

Classification Report: SVM				
	precision	recall	f1-score	support
birth_certificate_process	0.50	1.00	0.67	3.00
birth_certificate_requirements	0.75	1.00	0.86	3.00
business_permit_fee	1.00	1.00	1.00	1.00
cao_annotation	0.00	0.00	0.00	0.00
cao_certification	0.50	1.00	0.67	1.00
...	...	...	...	...
real_property_tax_due_date	1.00	1.00	1.00	3.00
real_property_tax_payment	0.67	0.67	0.67	3.00
accuracy	0.74	0.74	0.74	0.74
macro avg	0.52	0.57	0.53	95.00
weighted avg	0.69	0.74	0.70	95.00

61 rows x 4 columns

Classification Report: Logistic Regression				
	precision	recall	f1-score	support
birth_certificate_process	0.12	1.00	0.21	3.00
birth_certificate_requirements	0.67	0.67	0.67	3.00
business_permit_fee	0.00	0.00	0.00	1.00
cao_annotation	0.00	0.00	0.00	0.00
cao_certification	0.33	1.00	0.50	1.00
...	...	...	...	...
real_property_tax_due_date	1.00	1.00	1.00	3.00
real_property_tax_payment	1.00	0.67	0.80	3.00
accuracy	0.58	0.58	0.58	0.58
macro avg	0.35	0.37	0.34	95.00
weighted avg	0.60	0.58	0.56	95.00

61 rows x 4 columns

Classification Report: Neural Network (MLP)				
	precision	recall	f1-score	support
birth_certificate_process	0.67	0.67	0.67	3.00
birth_certificate_requirements	0.50	1.00	0.75	3.00
business_permit_fee	0.00	0.00	0.00	1.00
cao_annotation	0.00	0.00	0.00	0.00
cao_certification	0.33	1.00	0.50	1.00
...	...	...	...	...
real_property_tax_due_date	0.67	0.67	0.67	3.00
real_property_tax_payment	0.25	0.33	0.29	3.00
accuracy	0.69	0.69	0.69	0.69
macro avg	0.48	0.54	0.49	95.00
weighted avg	0.66	0.69	0.66	95.00

61 rows x 4 columns

Figure 23. Classification Report Results in Google Colab

To further analyze where the models succeeded or failed in predicting specific intents, confusion matrices were generated for each algorithm. Figure 24 and Figure 25 illustrate the confusion matrices for the Support Vector Machine (SVM) and Logistic Regression models, respectively. As seen in the visualizations, both models struggled significantly, displaying scattered predictions with high rates of false positives and false negatives. This indicates their inability to effectively separate the complex, multilingual intent categories present in the dataset.

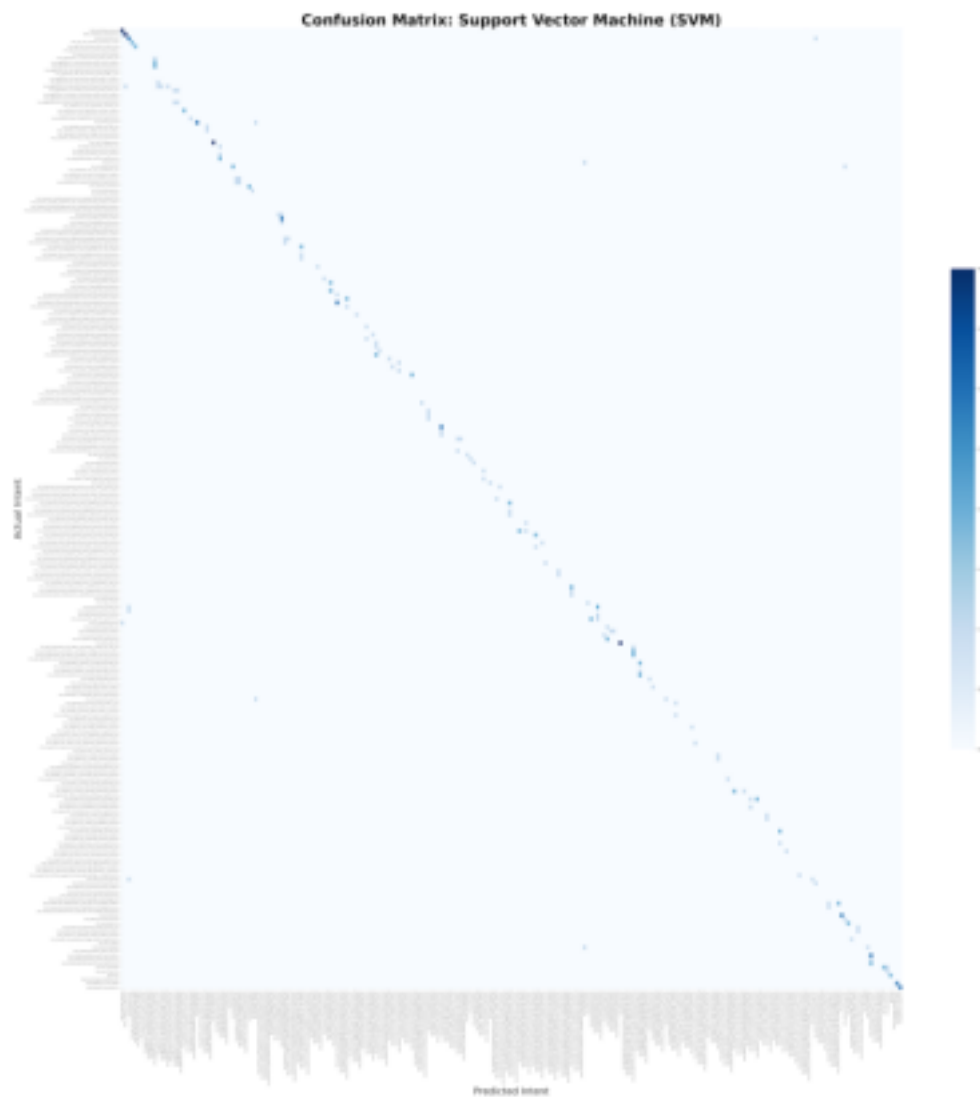


Figure 24. Confusion Matrix for SVM

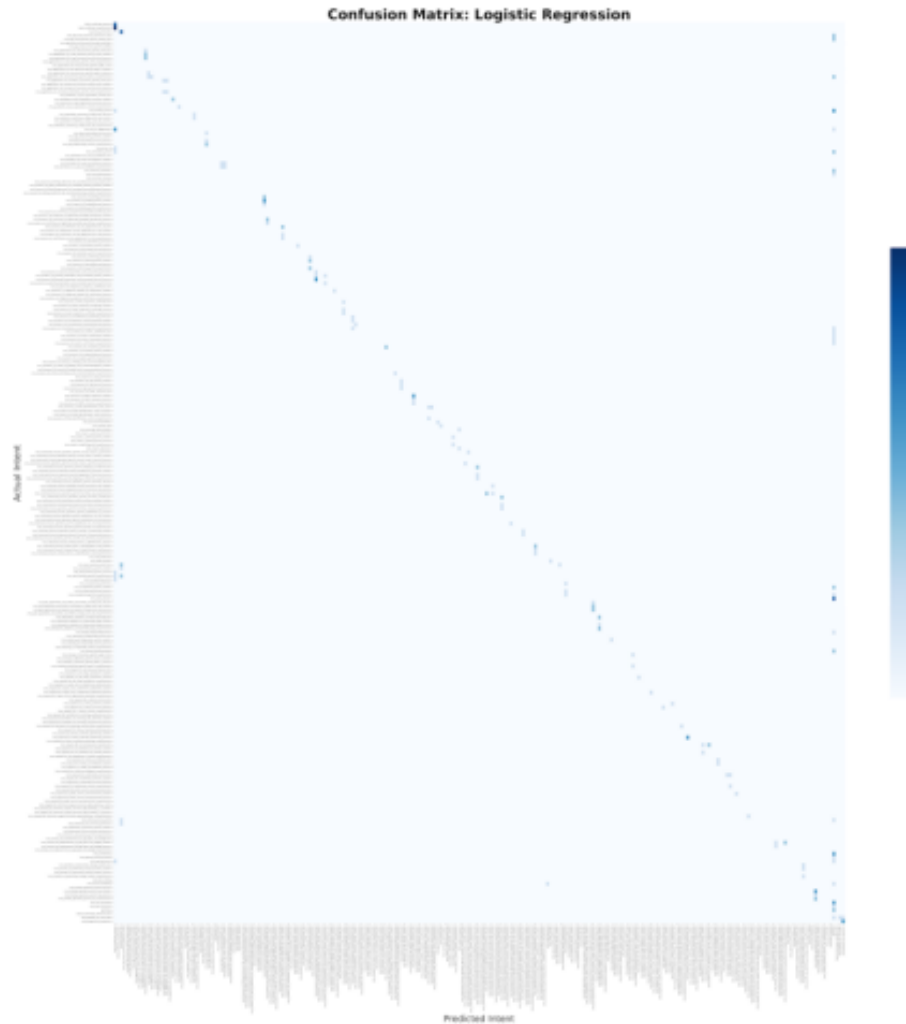


Figure 25. Confusion Matrix for Logistic Regression

In stark contrast, Figure 26 presents the confusion matrix for the Neural Network (MLP). The distinct, dark diagonal line across the matrix indicates a high volume of True Positives, confirming that the MLP model successfully learned the non-linear patterns of the textual data and made highly accurate predictions across almost all intent categories with minimal confusion.

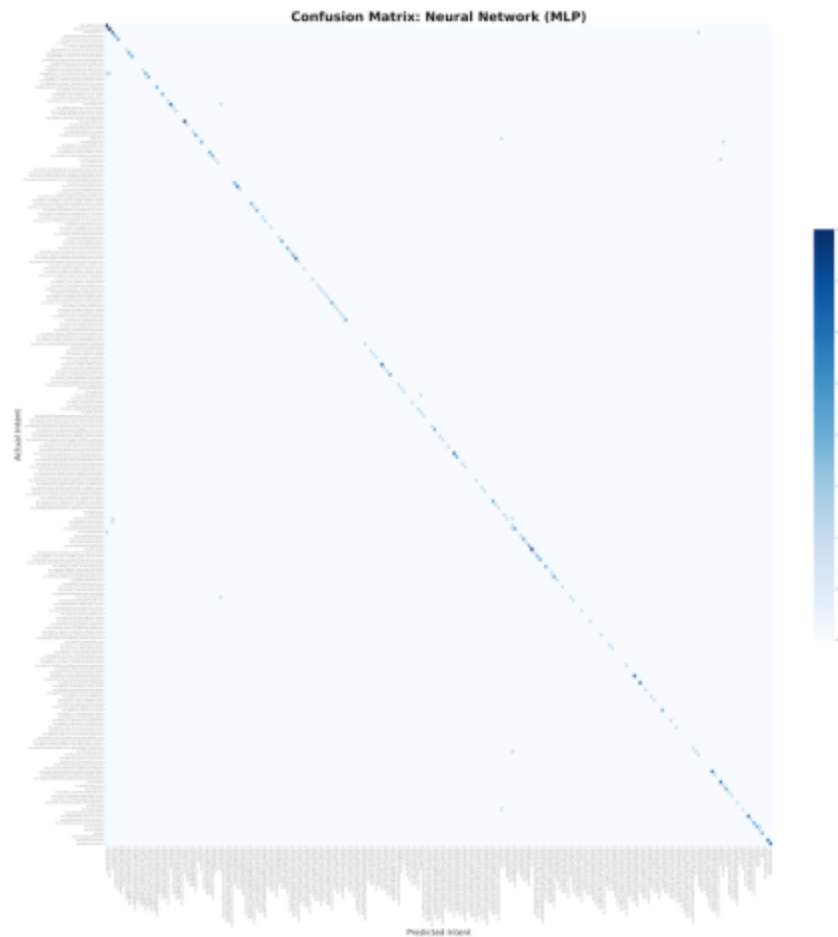


Figure 26. Confusion Matrix for Neural Network (MLP)

A direct visual comparison of the models' overall predictive accuracy is illustrated in the bar chart in Figure 27. This chart highlights the massive performance gap between the traditional linear models and the deep learning approach for this specific natural language dataset.

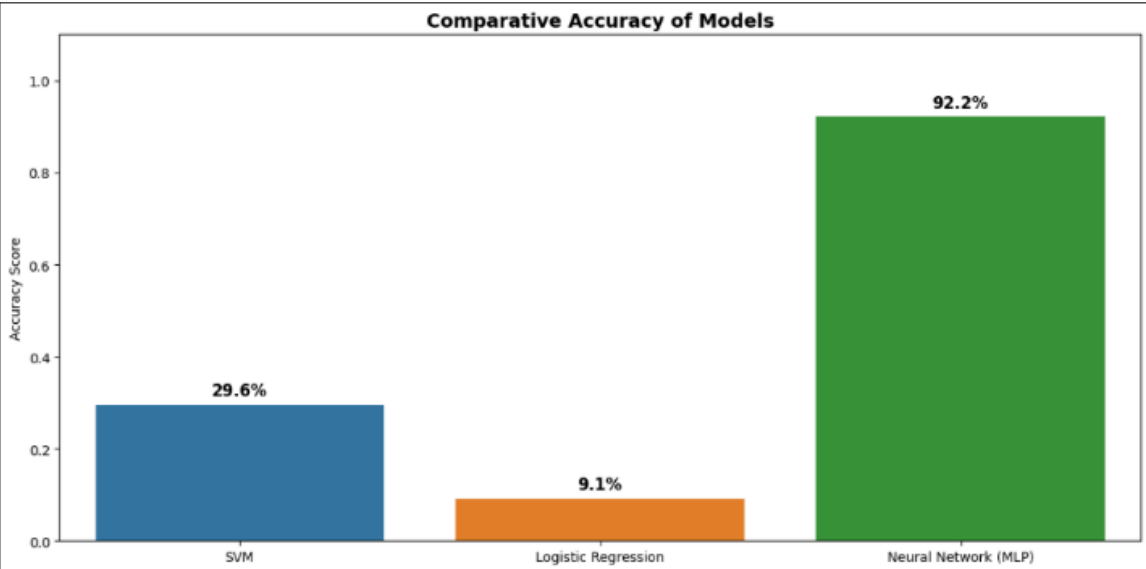


Figure 27. Comparative Accuracy of SVM, Logistic Regression, and MLP

The numerical breakdown of these results is presented in Table 3. As shown in the table, the Logistic Regression model performed the poorest, achieving an accuracy of only 0.09 (9%) and an F1-Score of 0.06, rendering it unusable for the system. The Support Vector Machine (SVM) performed marginally better but still yielded an unacceptable accuracy of 0.30 (30%). However, the Neural Network (MLP) demonstrated outstanding performance, achieving an accuracy of 0.92 (92%), a precision of 0.95 (95%), a recall of 0.92 (92%), and an F1-Score of 0.92.

Table 3. Comparative Evaluation Metrics of Three Classification Algorithms

Algorithm	Accuracy	Precision	Recall	F1-Score
Support Vector Machine (SVM)	0.30	0.30	0.30	0.29

Logistic Regression	0.09	0.05	0.09	0.06
Neural Network (MLP)	0.92	0.95	0.92	0.92

Based on the evaluation metrics detailed in Table 3 and the supporting visual figures, the Neural Network (MLP) was definitively selected as the core classification algorithm for the VISTA system. It proved to be the only model capable of reliably understanding and categorizing the complex, multilingual citizen queries with the high confidence required for deployment.

The following subsections describe the planned testing strategy and implementation approach for deploying VISTA in Panabo City Hall.

Systems Test Plan

The test plan encompasses three levels of testing: (1) Unit Testing of individual NLP pipeline components (preprocessor, vectorizer, classifier); (2) Integration Testing of the complete request-response flow from frontend to backend to database; and (3) User Acceptance Testing with actual LGU staff and citizens to validate usability and response quality.

Systems Implementation Plan

The implementation plan includes: (1) deployment of the backend server on the LGU's local infrastructure at Panabo City Hall or on a cloud hosting platform such as a virtual private server (VPS) to ensure 24/7 availability; (2) configuration of Firebase Authentication for administrator accounts; (3) initial population of the knowledge base from the Citizen's Charter with approximately 1,500 curated question variations across

five frontline offices; (4) training and evaluation of the three ML models in Google Colab to select the optimal classifier; and (5) a soft launch period with selected LGU offices in Panabo City before full public deployment. The system will be piloted and tested specifically within Panabo City, Davao del Norte, with frontline staff from the participating offices and selected citizen volunteers serving as the initial user base for User Acceptance Testing.

Systems Security Plan

The VISTA system will implement security measures to protect citizen inquiry data, knowledge base content, and administrator access. Authentication will be required before the administrator can access the dashboard and manage system functions such as knowledge base editing, unmatched query review, and model retraining. Administrator credentials will be protected through Firebase Authentication, which handles password hashing, session token management, and secure credential storage.

Access control will be applied to ensure that only authorized LGU administrators can manage knowledge base intents, view inquiry analytics, review unmatched queries, and trigger model retraining. The system will also protect stored data in the SQLite database, including citizen chat logs, intent definitions, training data, and unmatched query records.

The system will maintain data integrity by storing records properly and preventing unauthorized modification of important system information. Input sanitization will be implemented to prevent injection attacks on the backend API. Rate limiting will be applied to prevent API abuse. The system will also monitor login activity and failed login attempts to help identify possible security issues. These security measures will help

ensure that VISTA remains reliable, protected, and suitable for handling citizen inquiries at Panabo City Hall.

Systems Maintenance Plan

The VISTA system will follow a structured maintenance plan to ensure continuous operation, accuracy, and reliability. Regular maintenance will include monthly reviews of the knowledge base to ensure alignment with the current Citizen's Charter, quarterly model retraining cycles incorporating accumulated administrator-validated queries, and weekly automated backups of the SQLite database to prevent data loss.

The administrator or system maintainer will also monitor classification accuracy trends through the analytics dashboard to determine when the model requires retraining. If a significant drop in accuracy is observed, or if new service categories are added by Panabo City Hall, the administrator will trigger the controlled retraining process to update the machine learning models.

Software maintenance will include updating the web application frontend, improving system functions based on user feedback, reviewing SQLite database records, fixing detected errors, and updating system documentation. The knowledge base responses, intent categories, and office directory information will also be reviewed regularly to ensure that the system remains aligned with actual Panabo City Hall services and processes. Through this maintenance plan, VISTA will continue to support accurate multilingual citizen inquiry classification, knowledge base management, and real-time chatbot response delivery.

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